

Deep Generative Architectures

Outline

1. Discriminative vs Generative Models

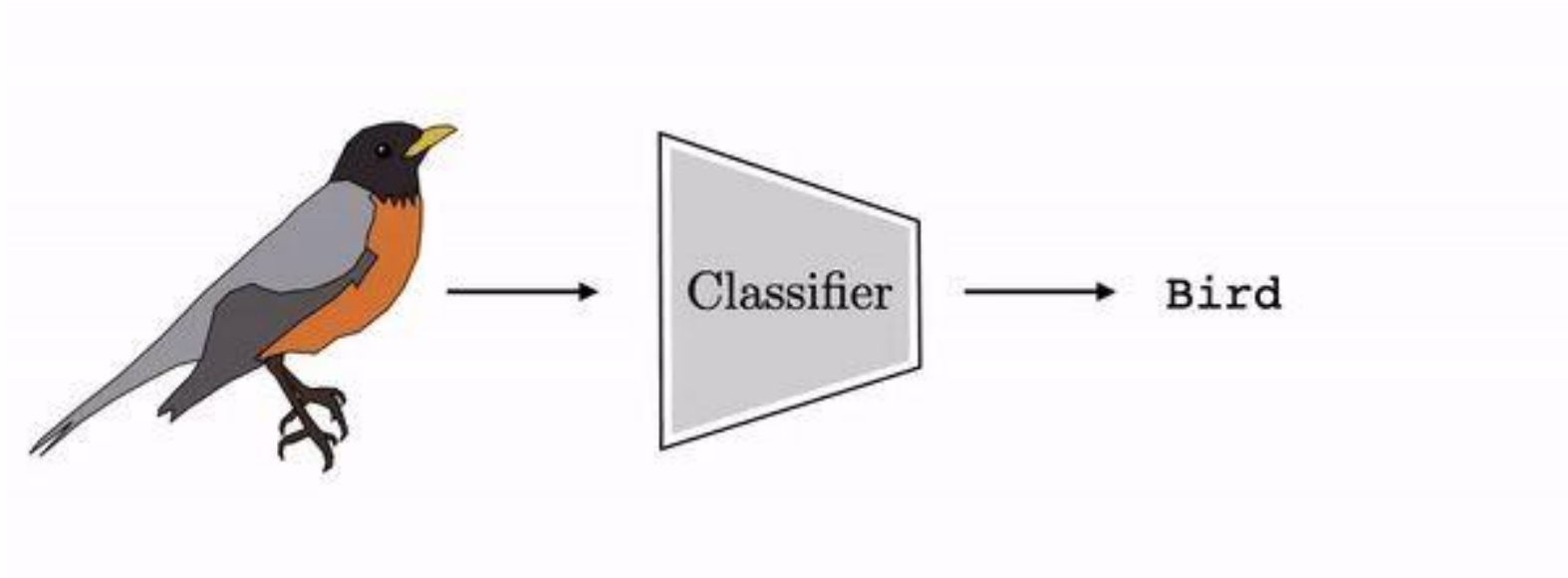
- a. $P(Y|x)$: Discriminative Models
- b. $P(x)$: Generative Models
- c. $P(x|Y)$: Conditioned Generative Models

2. Latent variable

3. Architectures

- a. GAN
- b. Auto-regressive
- c. VAE
- d. Diffusion

Discriminative vs Generative Models

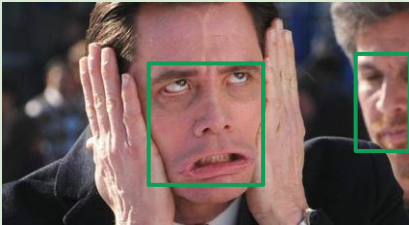


Outline

1. Discriminative vs Generative Models
 - a. **$P_{\theta}(Y|X)$: Discriminative Models**
 - b. $P_{\theta}(x)$: Generative Models
 - c. $P_{\theta}(x|Y)$: Conditioned Generative Models
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$P_{\theta}(Y|x)$: Discriminative Models

Slide credit:
Albert Pumarola (UPC 2019)

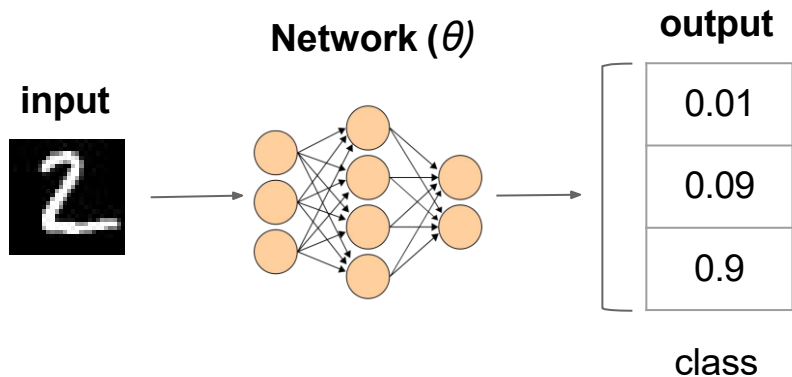
	Classification	Regression
Text		Prob. of being a Potential Customer
Image	 Jim Carrey	
Audio	 What Language?	Speech Translation

Discriminative Modeling
 $P_{\theta}(Y|x)$

x =Data
 Y =Labels
 θ = Model parameters

$P_{\theta}(Y|x)$: Discriminative Models

Discriminative model: Tell me the probability of some 'Y' responses given 'x' inputs.



$$P_{\theta}(Y \mid X = [\text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_{784}])$$

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1. Discriminative vs Generative Models

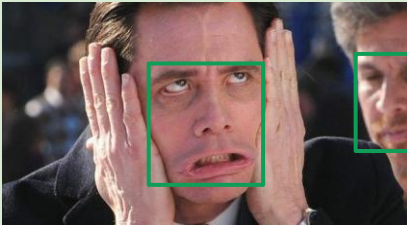
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- b. $P(X)$: Generative Models**
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2. Sampling

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$P_{\theta}(x)$: Generative Models

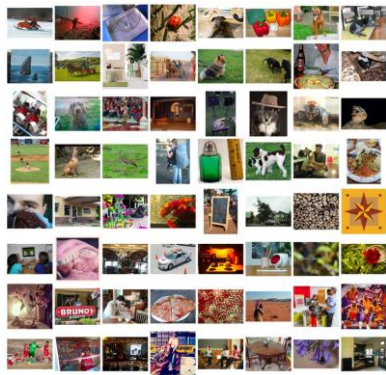
	Classification	Regression	Generative
Text		Prob. of being a Potential Customer	"What about Ron magic?" offered Ron. To Harry, Ron was loud, slow and soft bird. Harry did not like to think about birds.
Image	 Jim Carrey		
Audio	 What Language?	Language Translation	Music Composer and Interpreter MuseNet Sample

Discriminative Modeling
 $P_{\theta}(Y|x)$

Generative Modeling
 $P_{\theta}(x)$

x =Data
 Y =Labels
 θ = Model parameters

$P_{\theta}(x)$: Generative Models



Each real sample x_i comes from an M -dimensional probability distribution $P(X)$.

$$\mathbf{X} = \{x_1, x_2, \dots, x_N\}$$

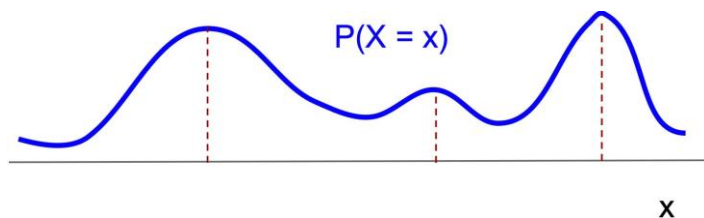
$P_{\theta}(x)$: Generative Models

- 1) We want our model with parameters θ to output samples with distribution $P_{\theta}(X)$, matching the distribution of our training data $P(X)$.
- 2) We can sample points from $P_{\theta}(X)$ plausibly looking how $P(X)$ distributed.

Example: Gaussian Mixture Models (GMM)

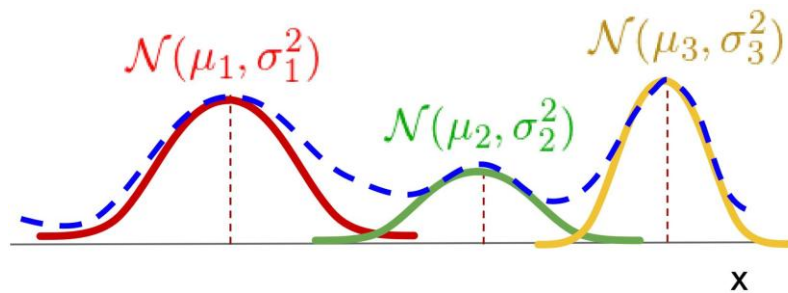
$P(X)$

Distribution of training data



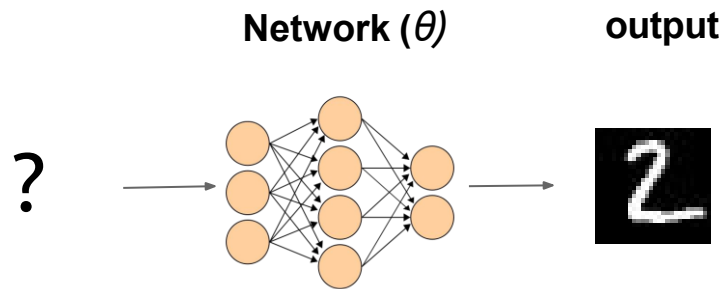
$P_{\lambda, \mu, \sigma}(X)$

Distribution of training data



$$P(X) = \lambda_1 \mathcal{N}(\mu_1, \sigma_1^2) + \lambda_2 \mathcal{N}(\mu_2, \sigma_2^2) + \dots$$

$P_{\theta}(x)$: Generative Models



What are the parameters θ we need to estimate in deep neural networks?

$\theta = (\text{weights \& biases})$

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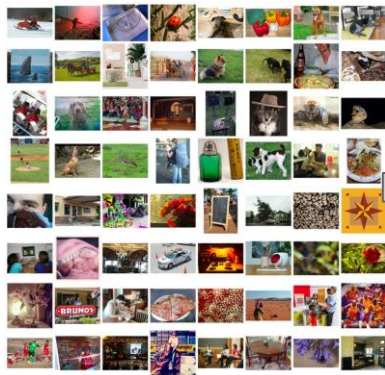
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- c. **$P(X|Y)$: Conditioned Generative Models**

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$P_{\theta}(x|Y)$: Conditioned Generative Models



DOG

CAT

TRUCK

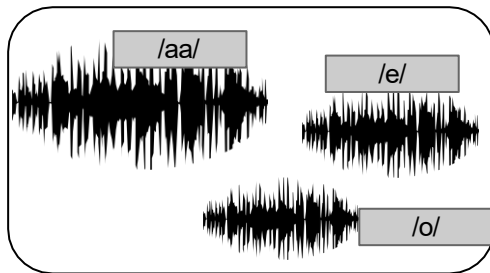
PIZZA



THRILLER

SCI-FI

HISTORY



Joint probabilities $P(X/Y)$ to model conditioning variables on the generative process:

$$\mathbf{X} = \{x_1, x_2, \dots, x_N\}$$

$$\mathbf{Y} = \{y_1, y_2, \dots, y_N\}$$

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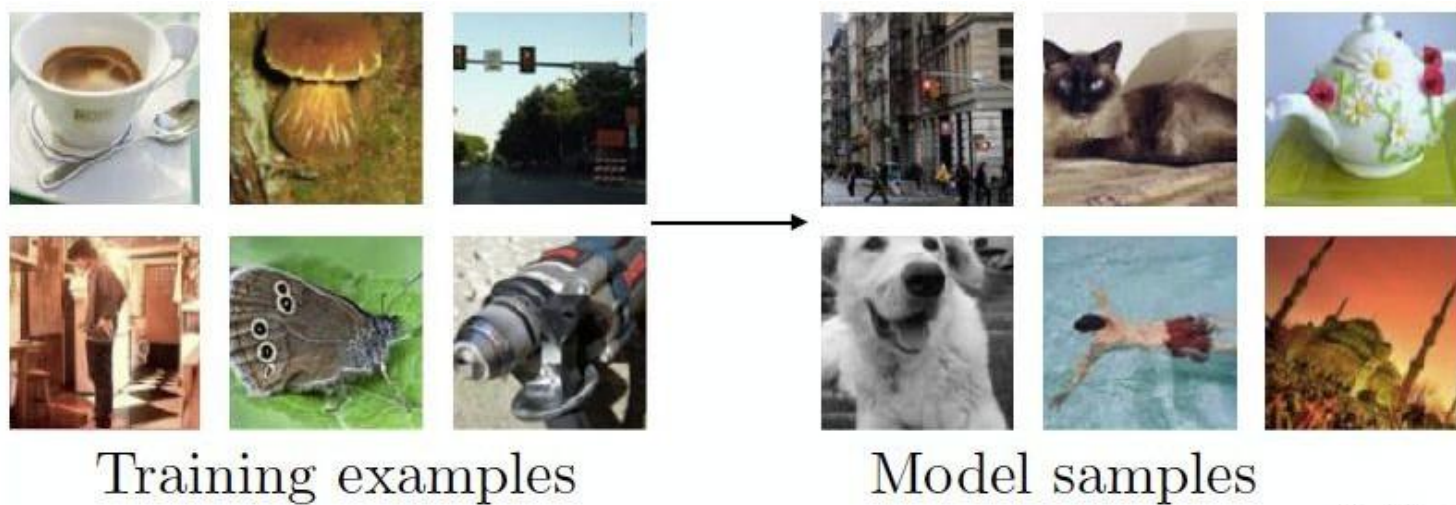
2. Sampling

3. Architectures

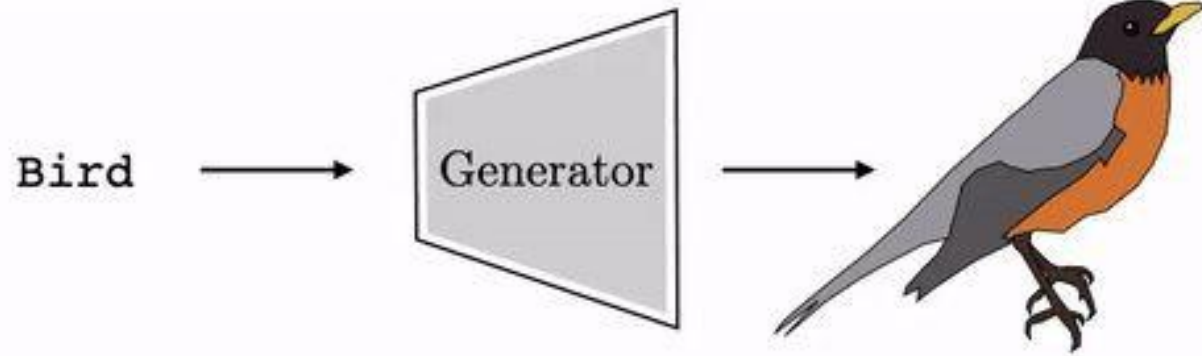
- a. Generative Adversarial Networks (GANs)
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Sampling

Our learned model should be able to **make up new samples** from the distribution, **not just copy and paste** existing samples!

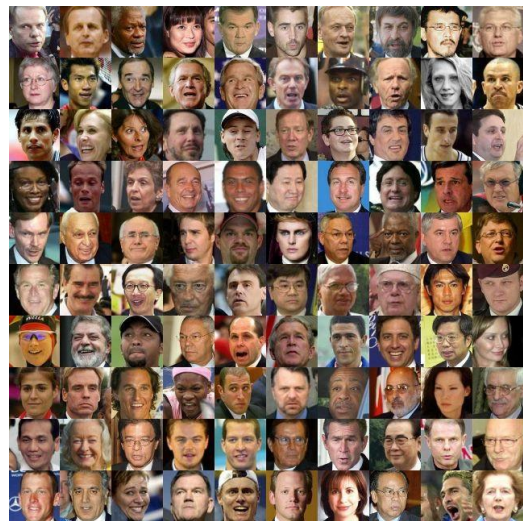


Sampling

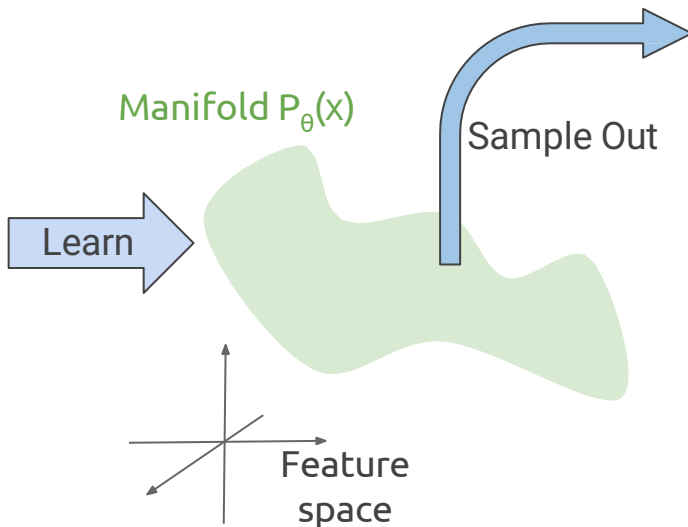


Sampling

“Model the data distribution so that we can sample new points out of the distribution”



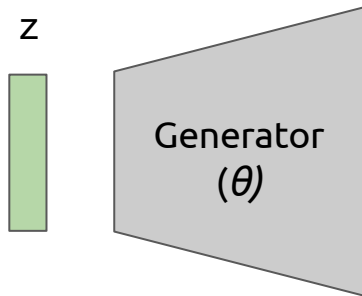
Training Dataset



Generated Samples

Sampling

How could we generate diverse samples from a deterministic deep neural network ?



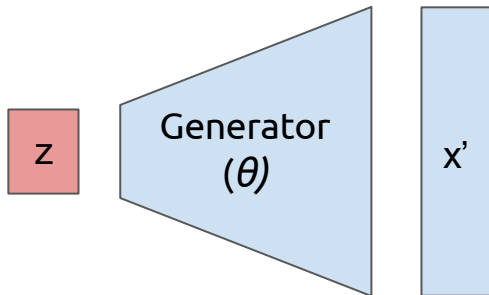
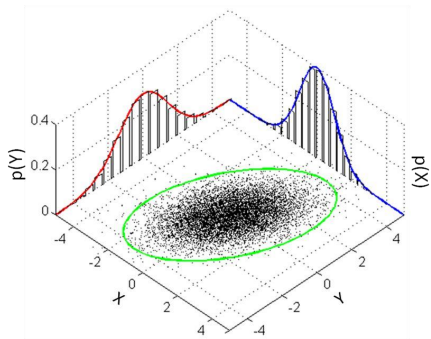
Generated Samples

Sampling

How could we generate diverse samples from a deterministic deep neural network ?

Sample z from a known prior, for example, a multivariate normal distribution $N(0, I)$.

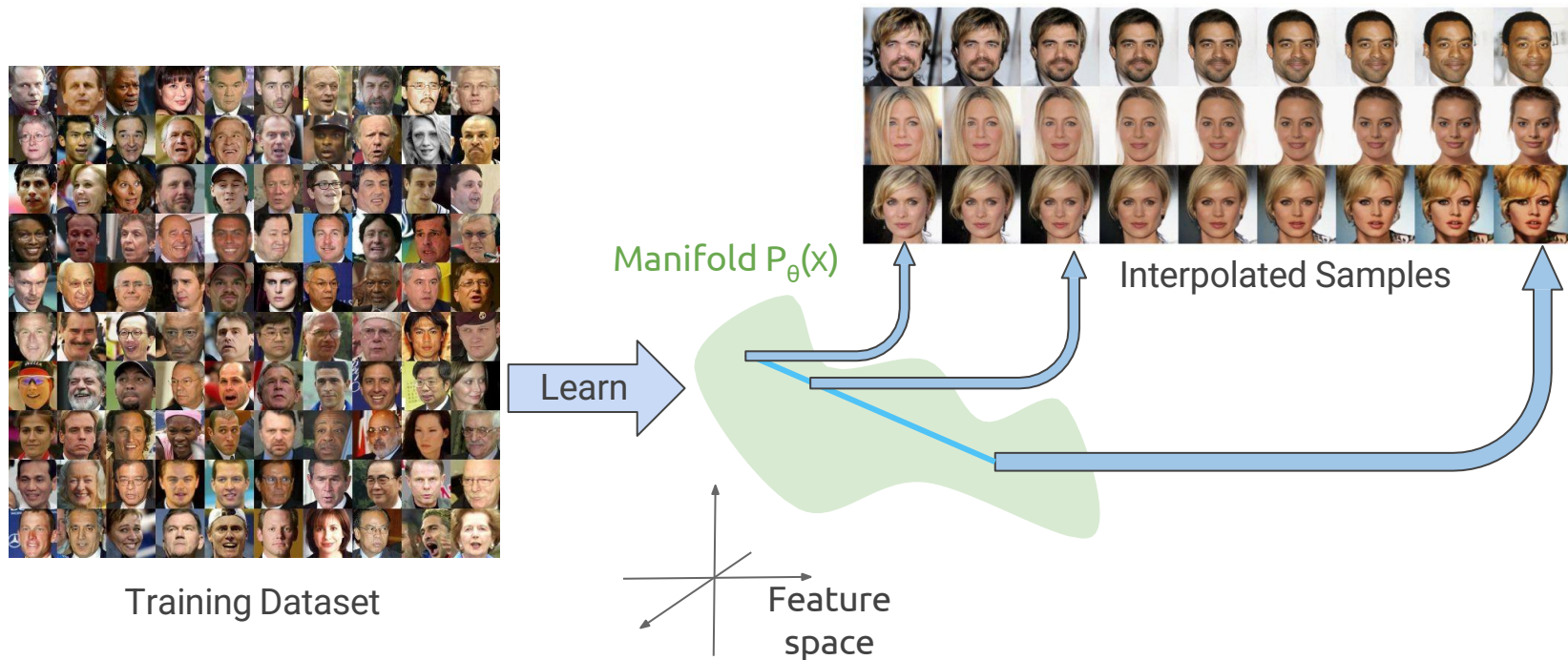
Example: $\dim(z)=2$



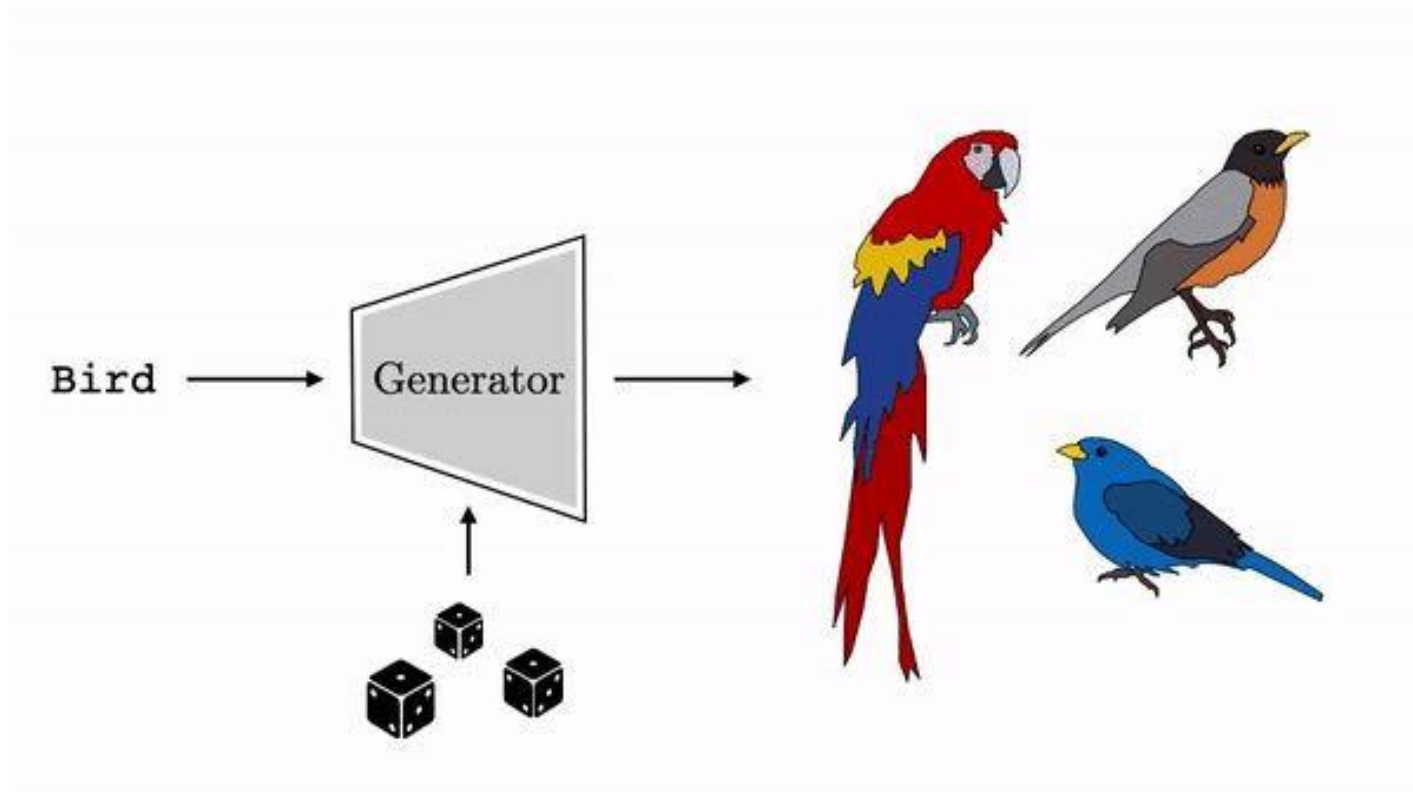
Generated Samples

Interpolation

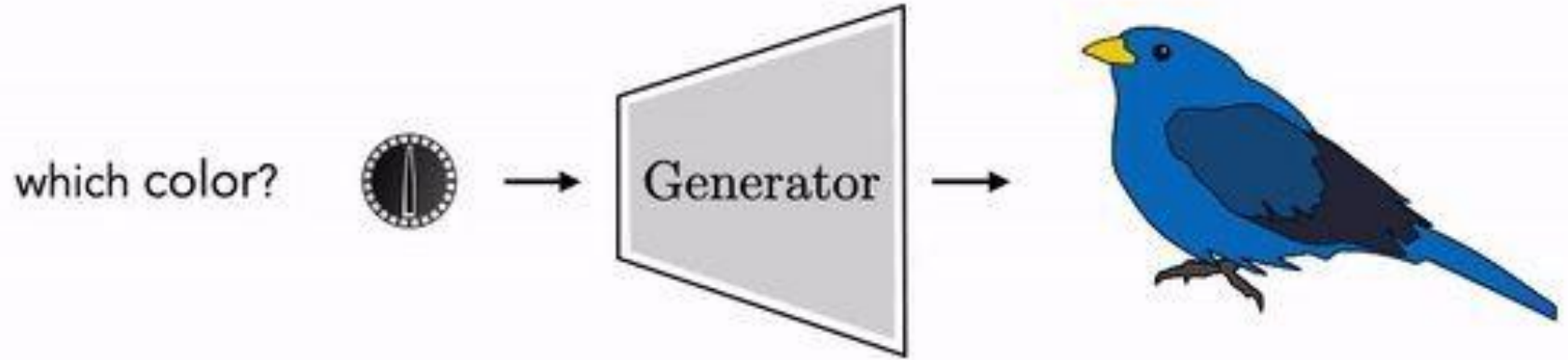
Traversing the learned manifold through interpolation.



Disentanglement



Disentanglement



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 - Generator & Discriminator Networks
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 - Diffusion

Generator & Discriminator

We have two modules: **Generator** (G) and **Discriminator** (D).

- They “fight” against each other during training→ **Adversarial Learning**

D's goal:

Classify between real samples and those produced by G.

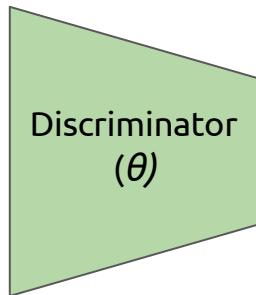
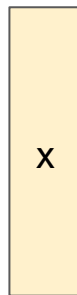
G's goal:

Fool D to misclassify.

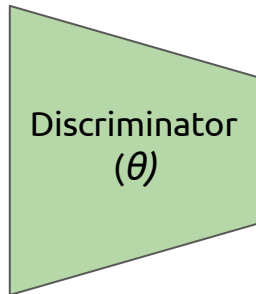
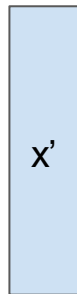


Discriminator

Discriminator network D $\rightarrow$ binary classifier between real (x) and generated (x') samples.



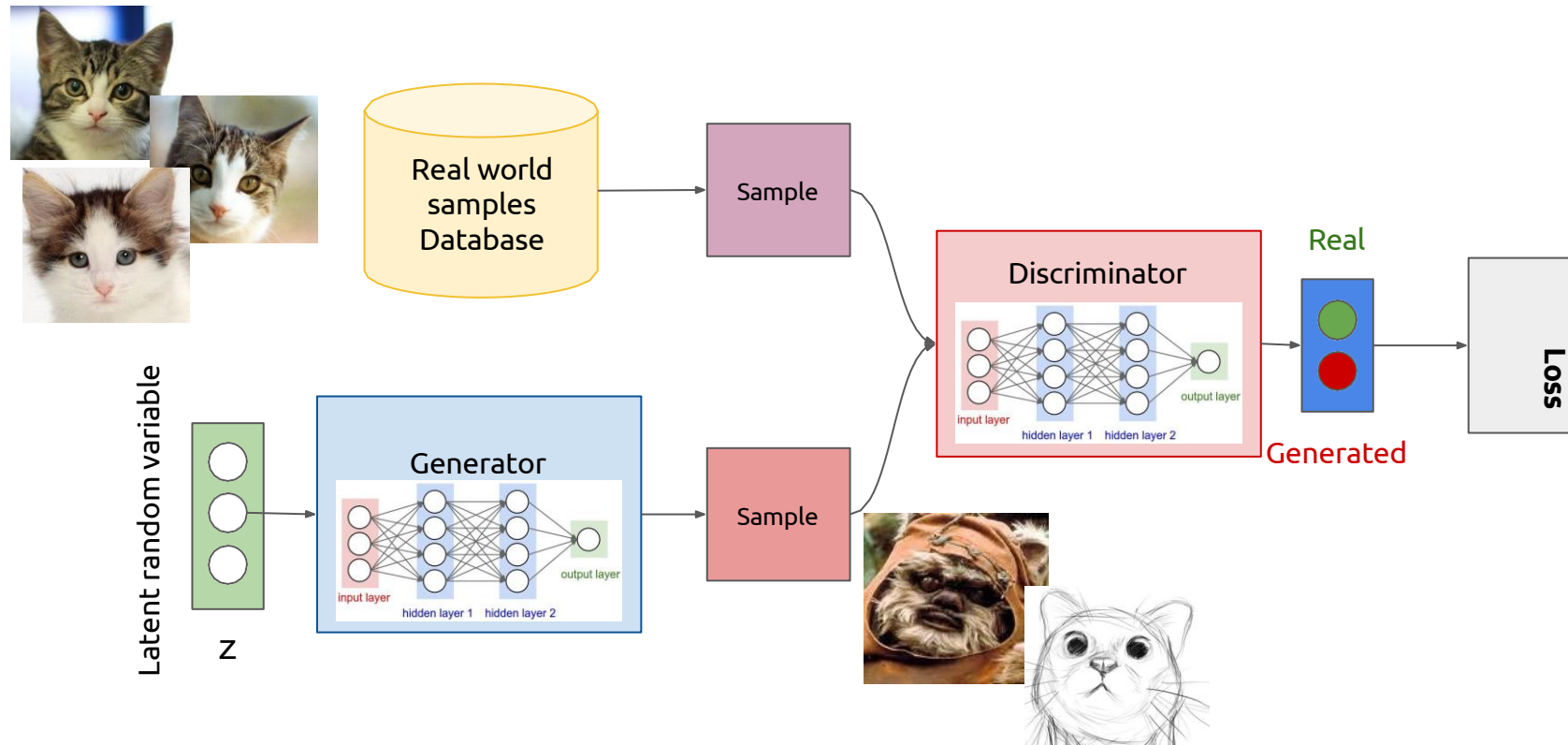
Real (0)



Generated (1)



Generator & Discriminator



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Adversarial Training Analogy: is it fake money?

Imagine we have a counterfeiter (**G**) trying to make fake money, and the police (**D**) has to detect whether money is real or fake.

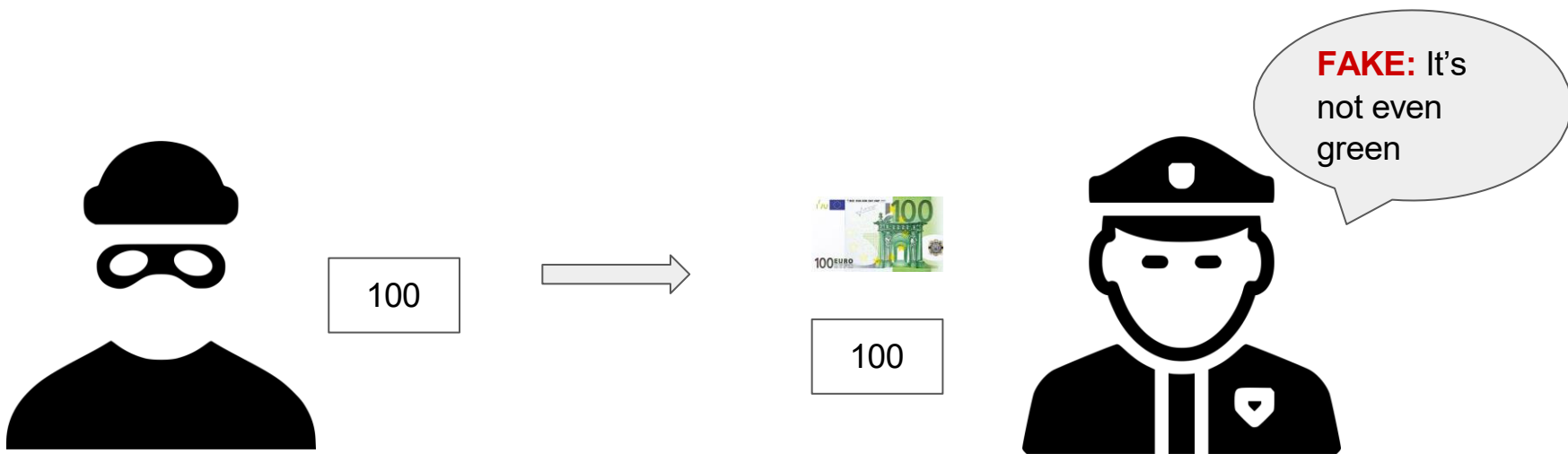


Figure: Santiago Pascual (UPC)

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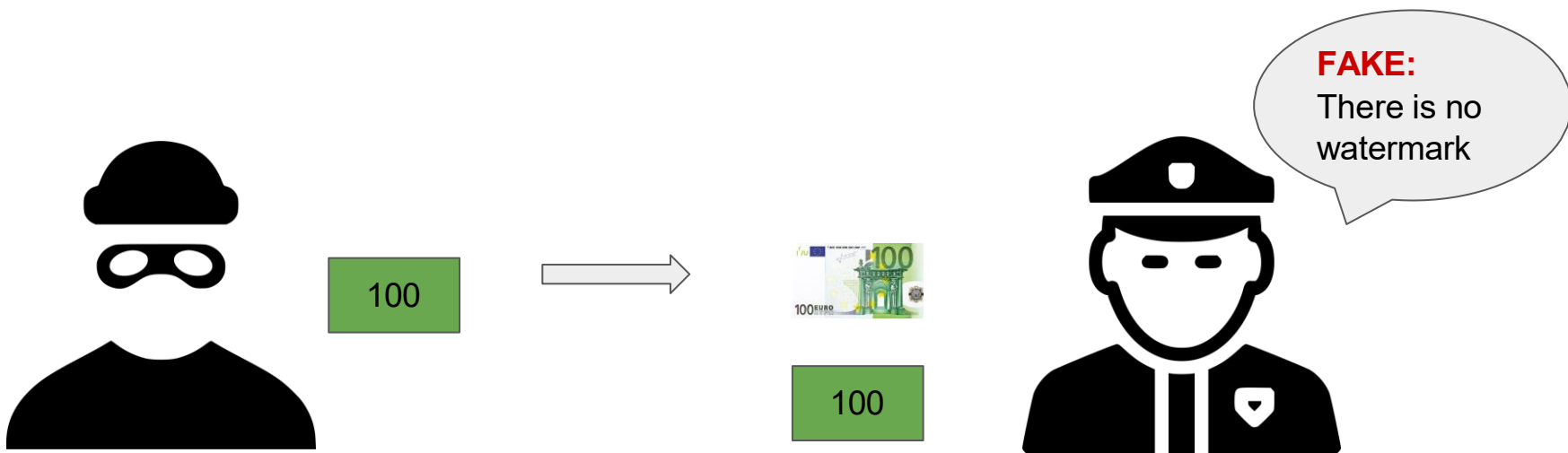


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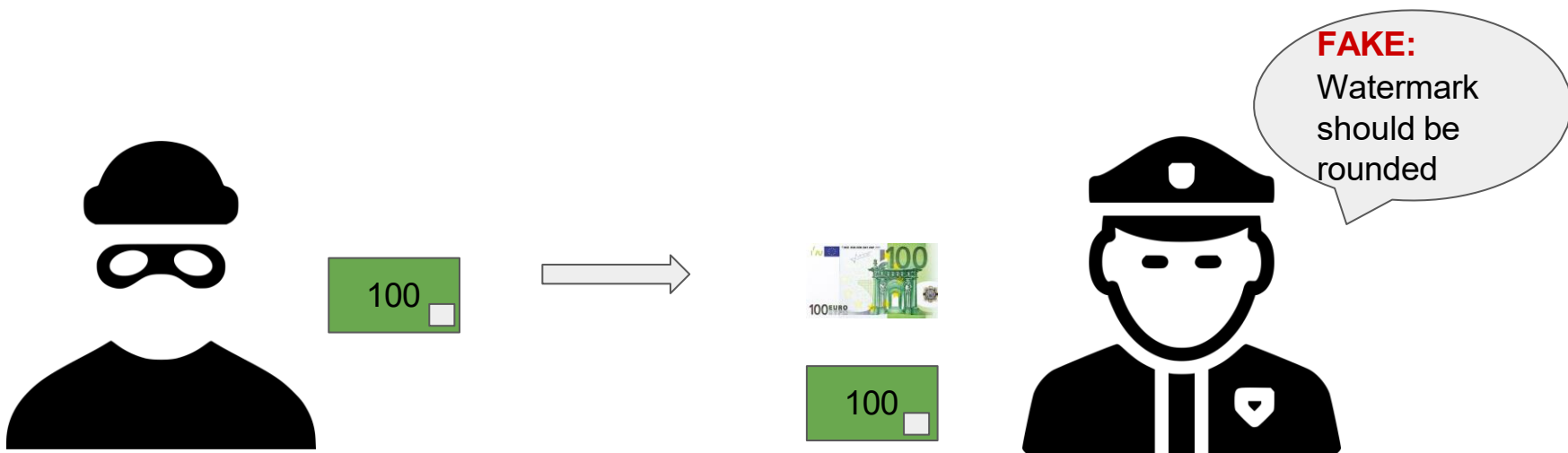
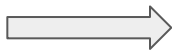


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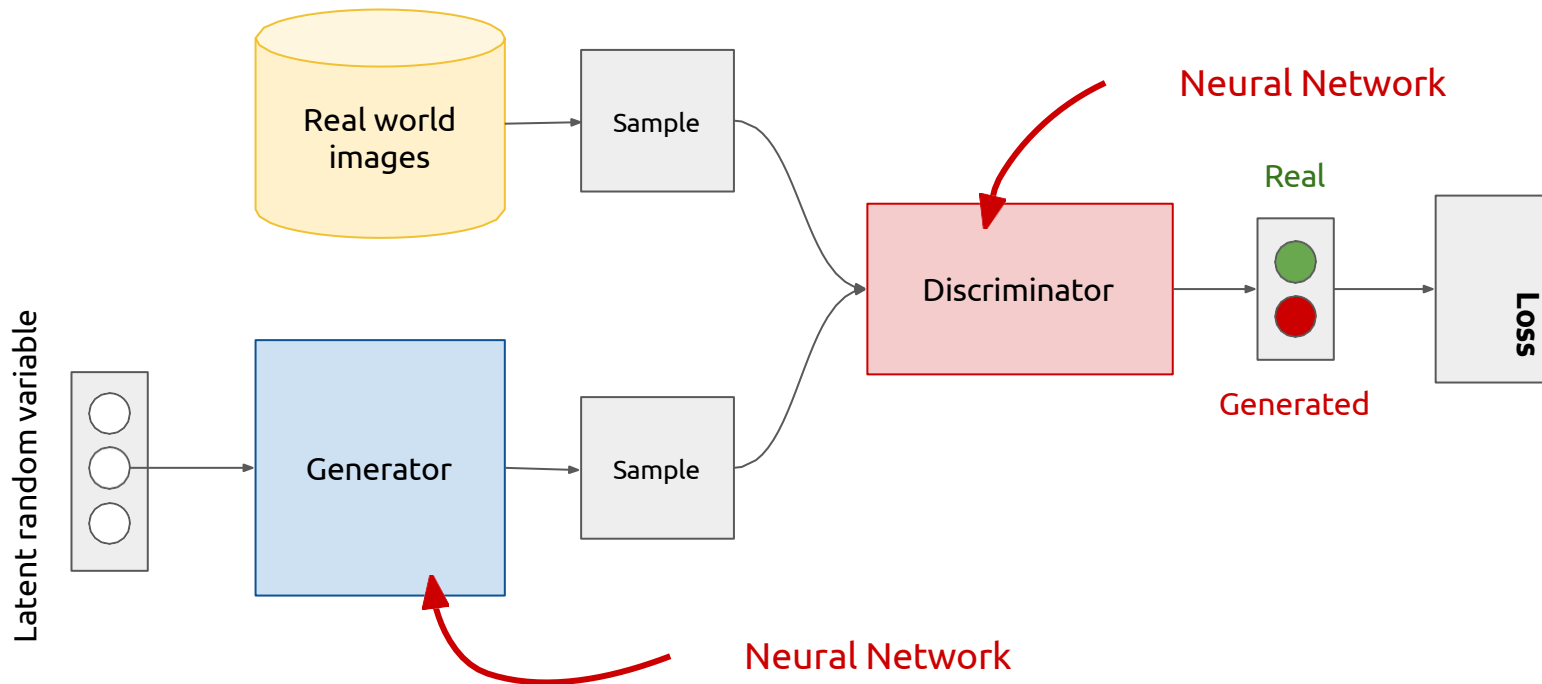
After enough iterations, and if the counterfeiter is good enough (in terms of **G** network it means "has enough parameters"), the police should be confused.



REAL?
FAKE?

Adversarial Training

Alternate between training the discriminator and generator



Adversarial Training: Discriminator

1. Fix generator weights, draw samples from both real world and generated images
2. Train discriminator to distinguish between real world and generated images

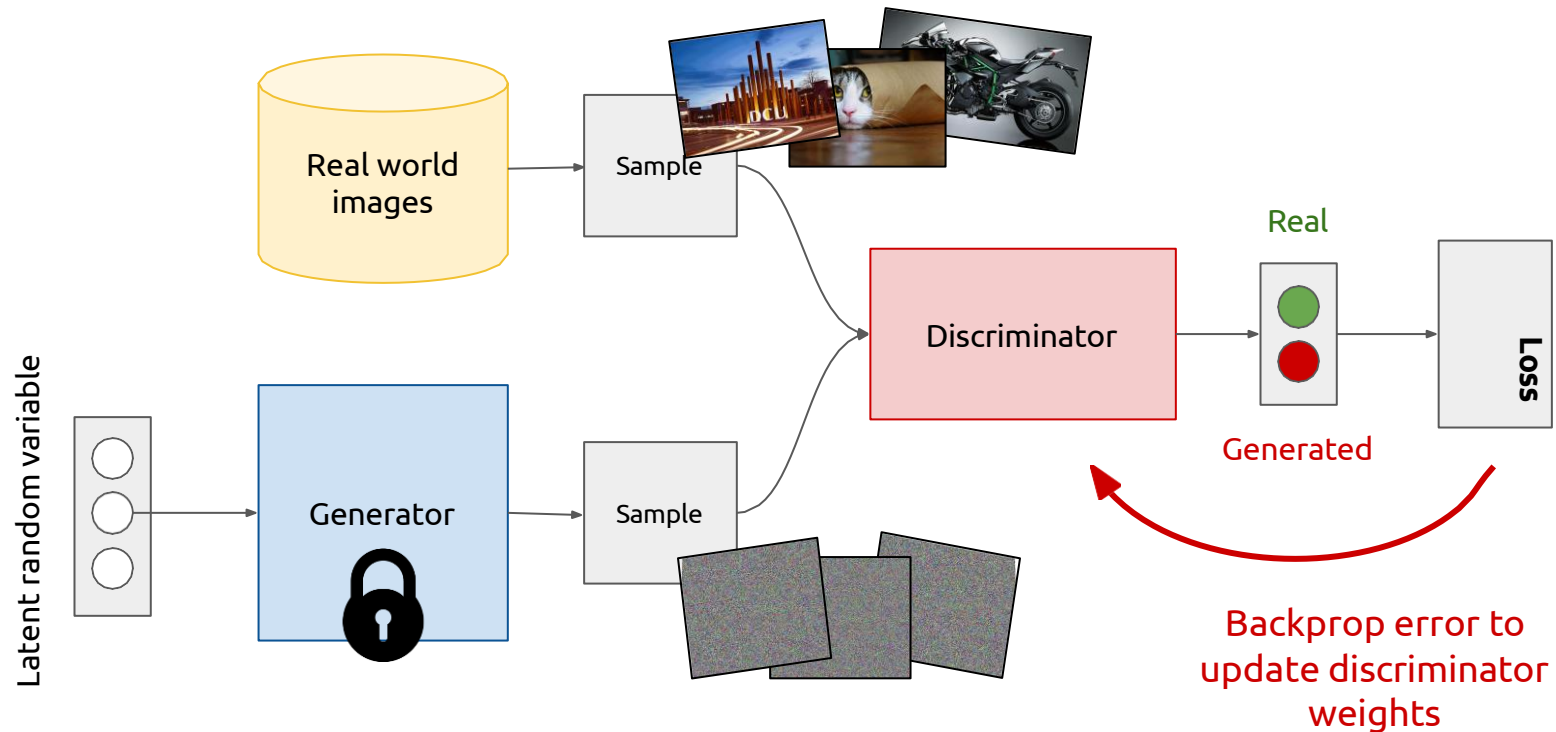


Figure: Kevin McGuinness (DCU)

Adversarial Training: Discriminator

In the set up of the figure, which ground truth label for a generated image should we use to train the discriminator ? Consider a binary encoding of "1" (Real) and "0" (Fake).

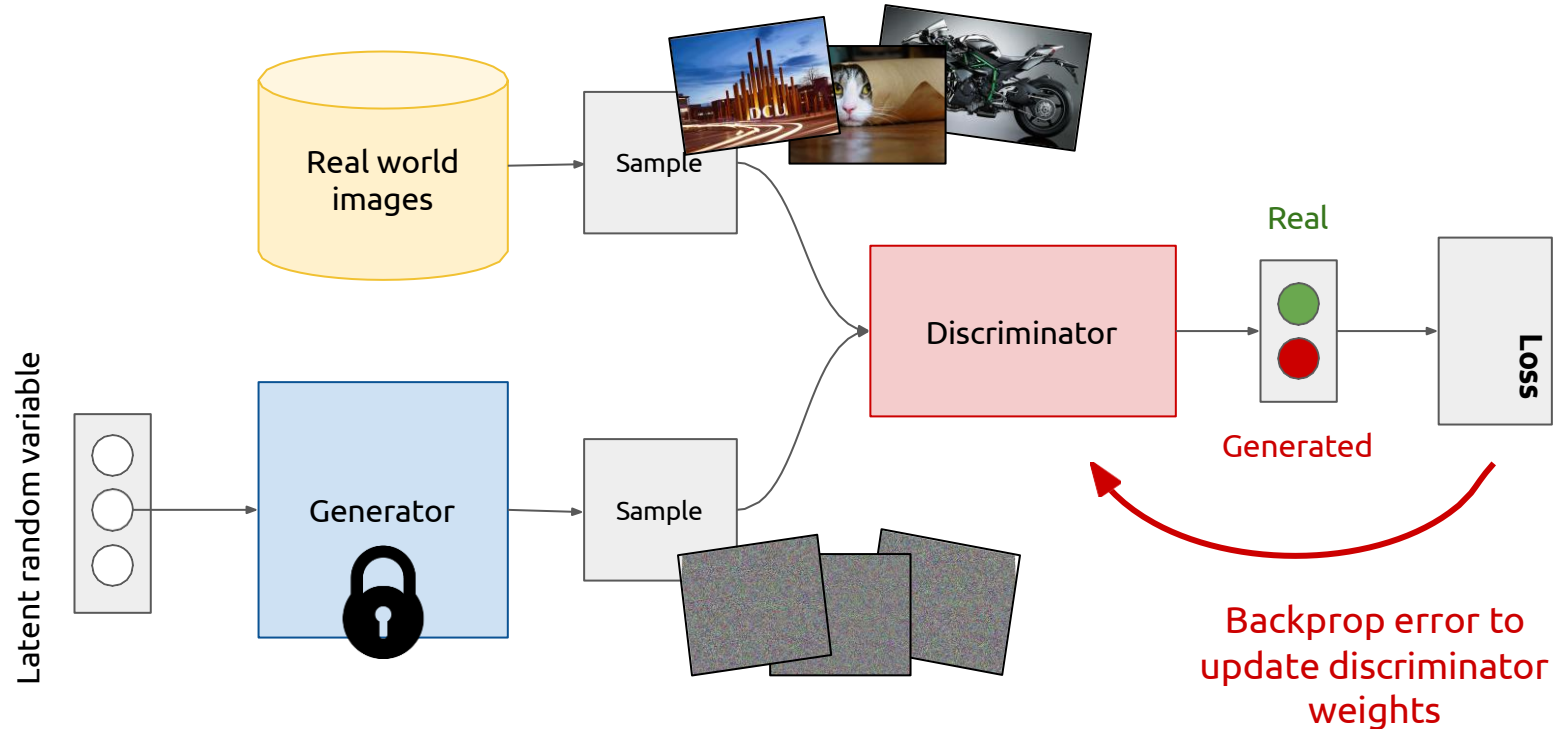


Figure: Kevin McGuinness (DCU)

Adversarial Training: Generator

1. Fix discriminator weights
2. Sample from generator by injecting noise.
3. Backprop error through discriminator to update generator weights

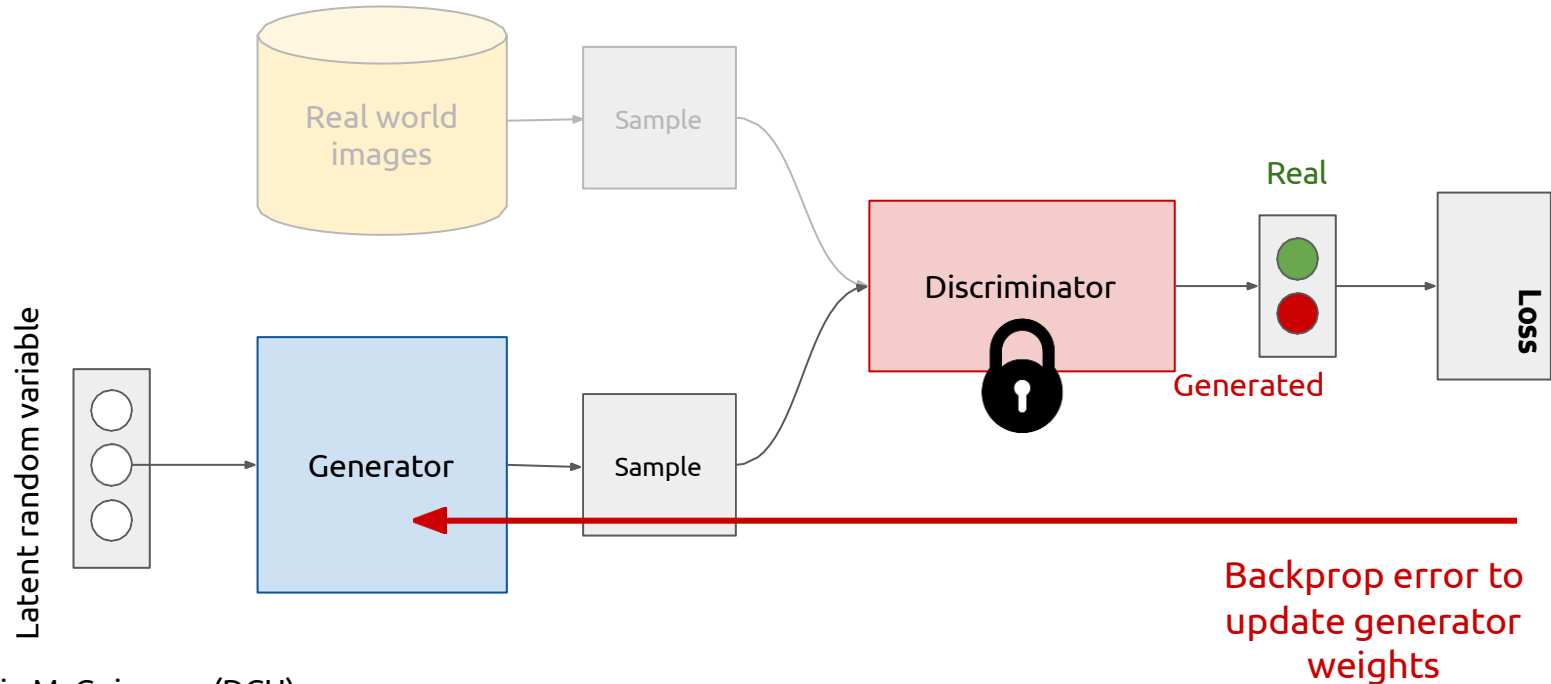


Figure: Kevin McGuinness (DCU)

Adversarial Training: Generator

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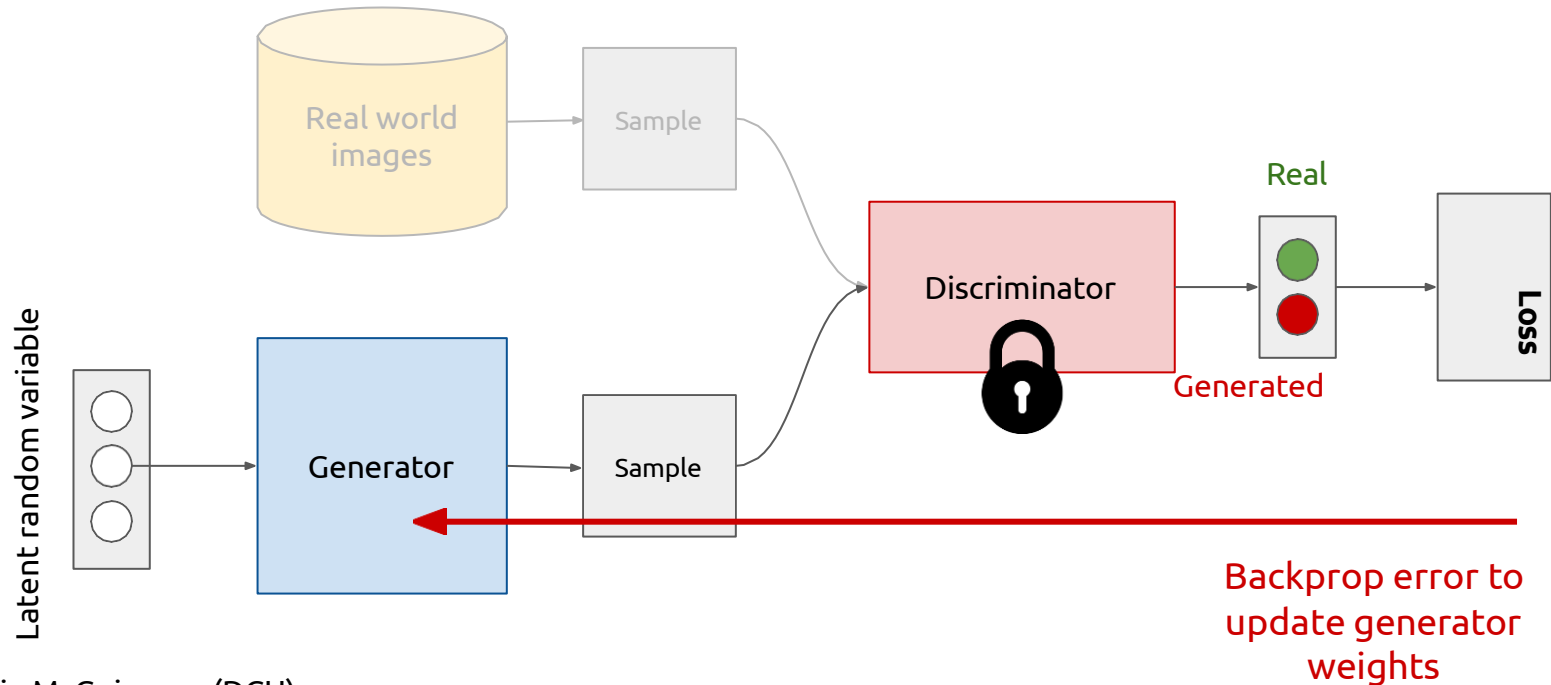


Figure: Kevin McGuinness (DCU)

Adversarial Training: How to make it work ?



Soumith Chintala
@soumithchintala

How to Train a GAN? I'm as clueless as you. Best I'll do is summarize the tricks, hacks and untold dark rituals that we've all used so far!

Tradueix el tuit



7:08 p. m. · 3 de des. de 2016 · Twitter Web Client



NeurIPS Barcelona 2016

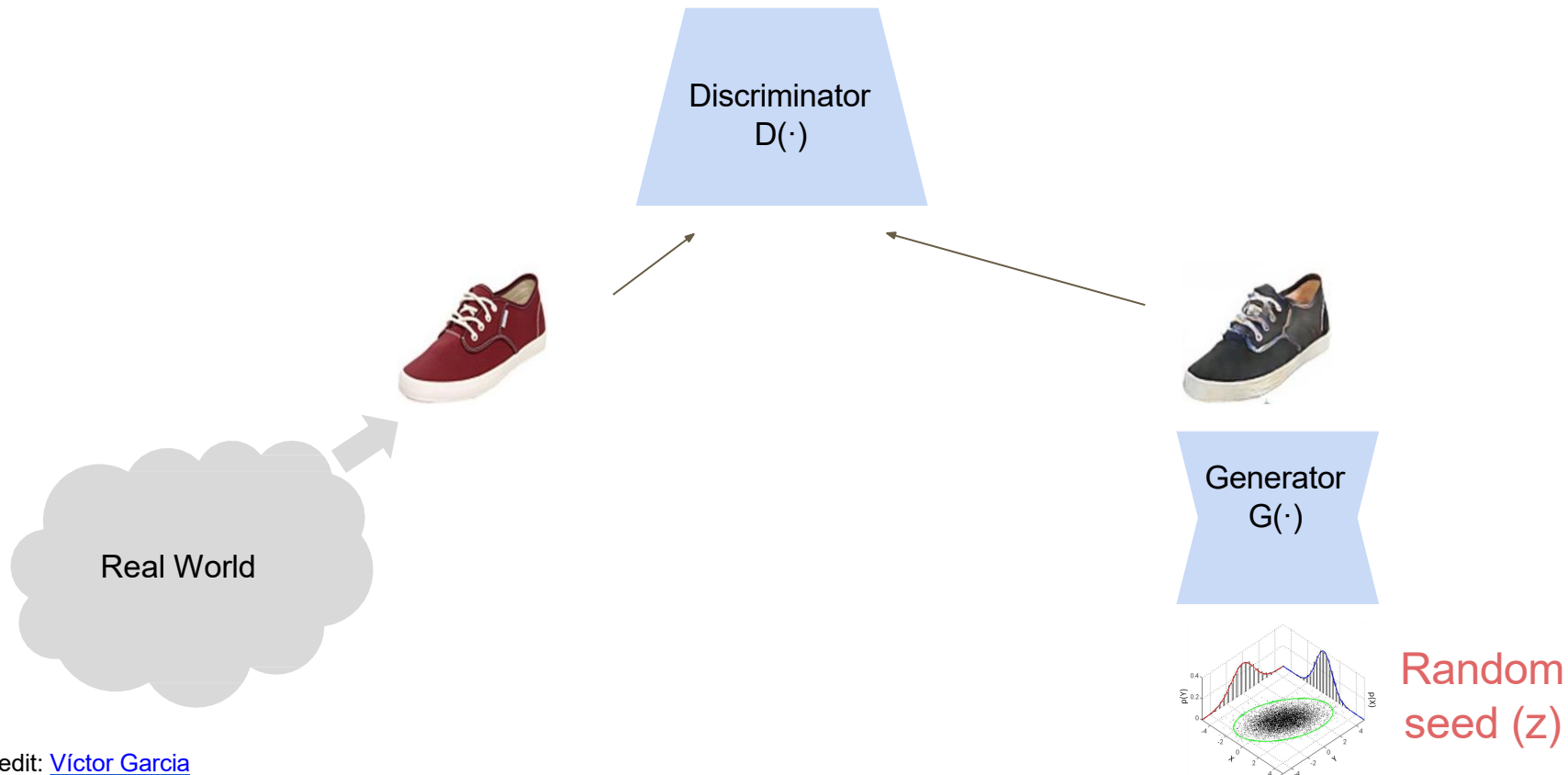
Soumith Chintala, [“How to train a GAN ? Tips and tricks to make GAN work”](#). Github 2016.

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Non-Conditional GANs

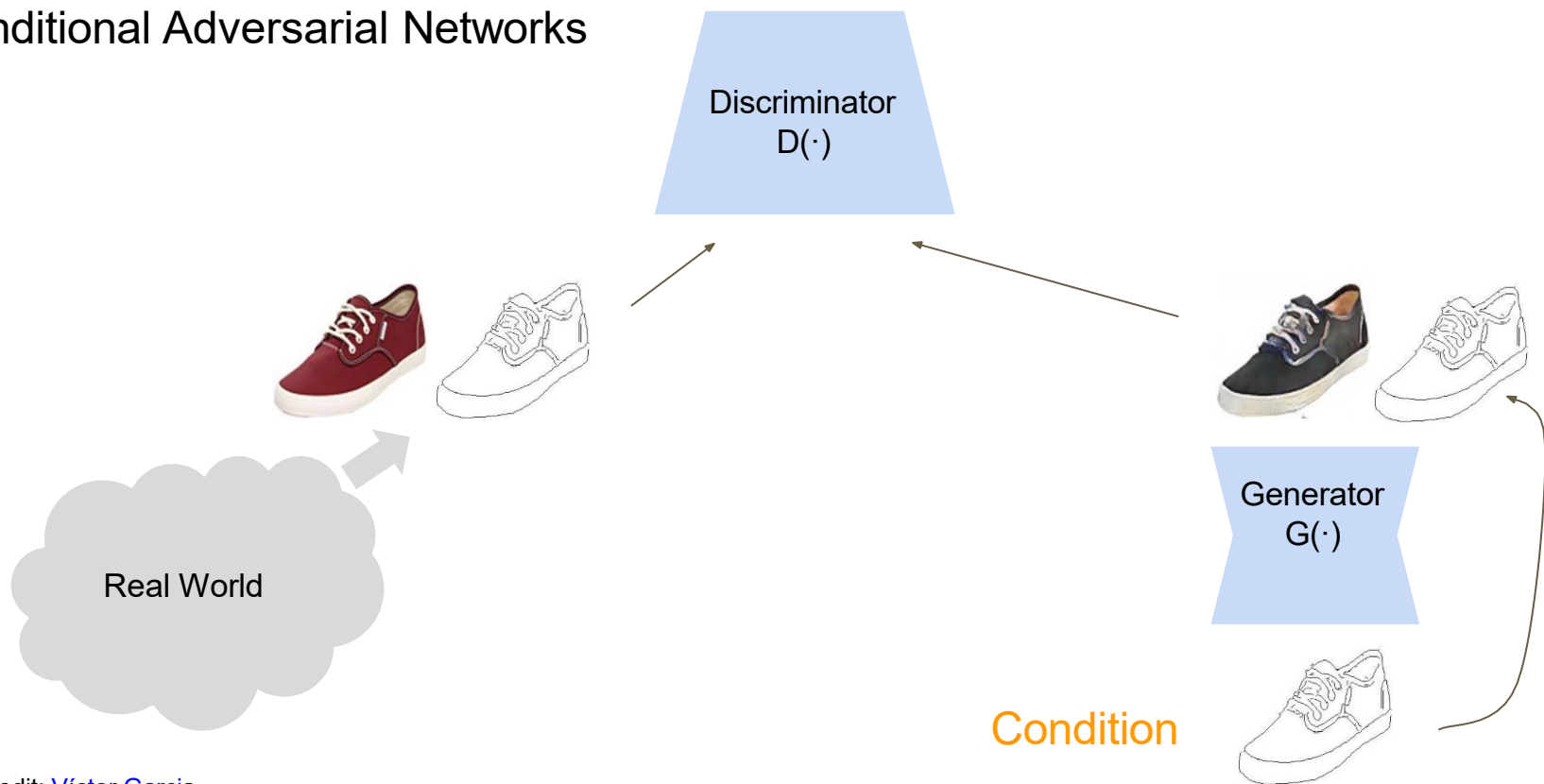
Real/Generated



Conditional GANs (cGAN)

Real/Generated

Conditional Adversarial Networks



Condition

Learn more about GANs



Ian Goodfellow.
NeurIPS Barcelona 2016.



Mihaela Rosca & Jeff Donahue.
[UCL x Deepmind 2020.](#)

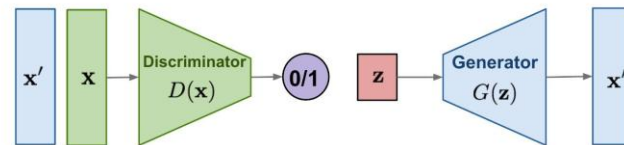
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GAN: Adversarial training

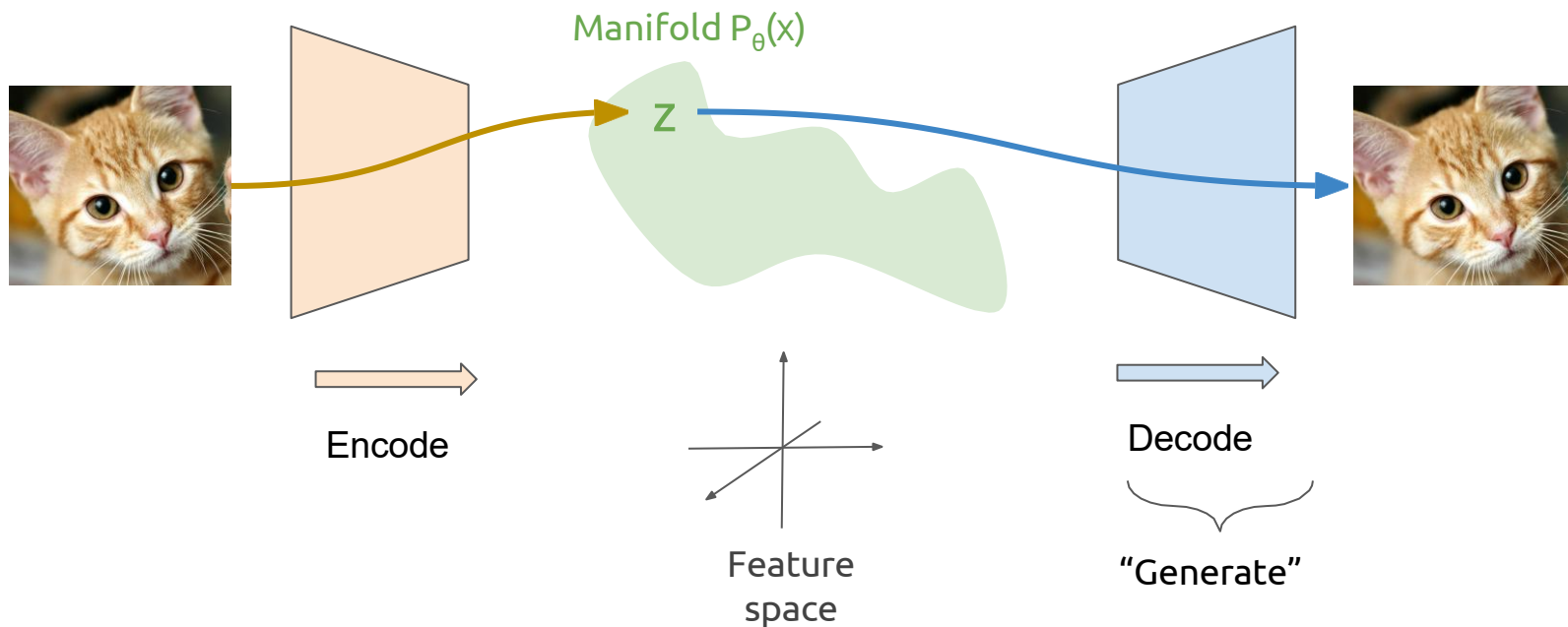


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 - Variational Inference
 - Reparametrization trick
 - Generative behaviour
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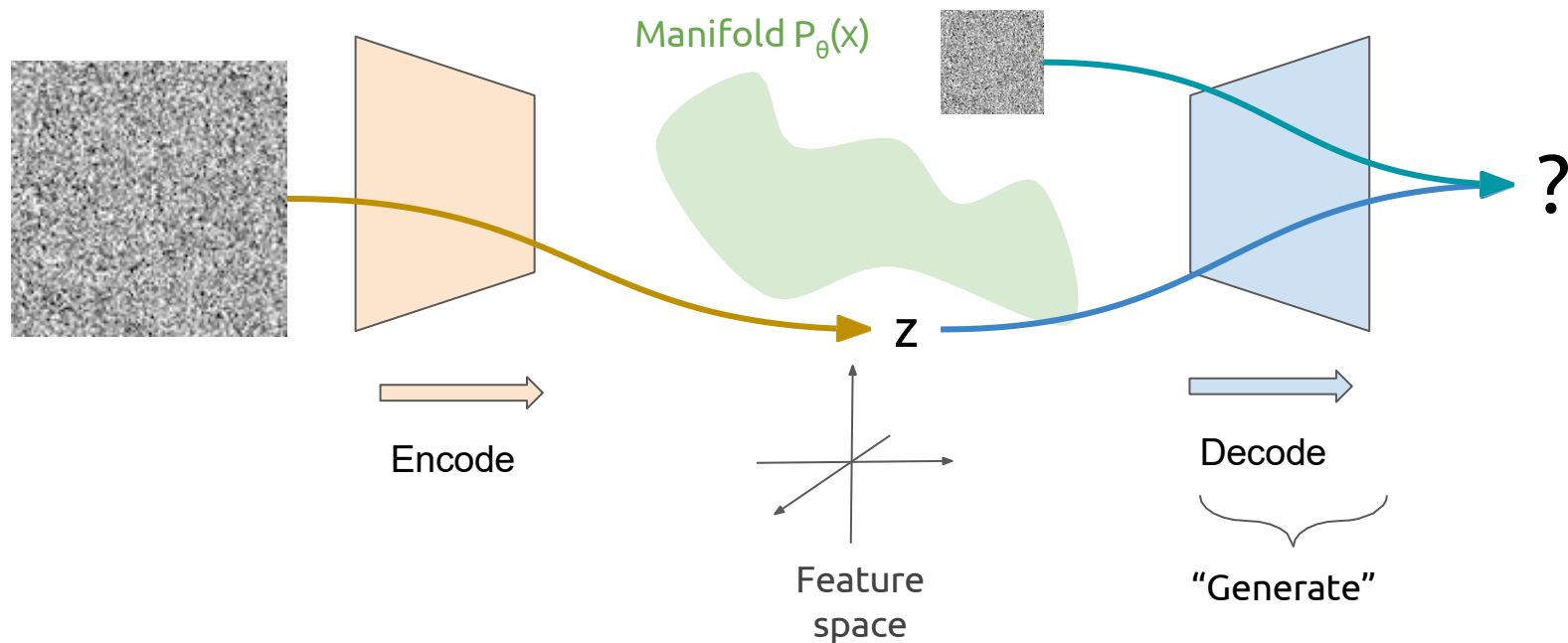
Auto-Encoder (AE)

- Learns $P_{\theta}(x)$ with a reconstruction loss.
- Proposed as a pre-training stage for the encoder (“self-supervised learning”).



Auto-Encoder (AE)

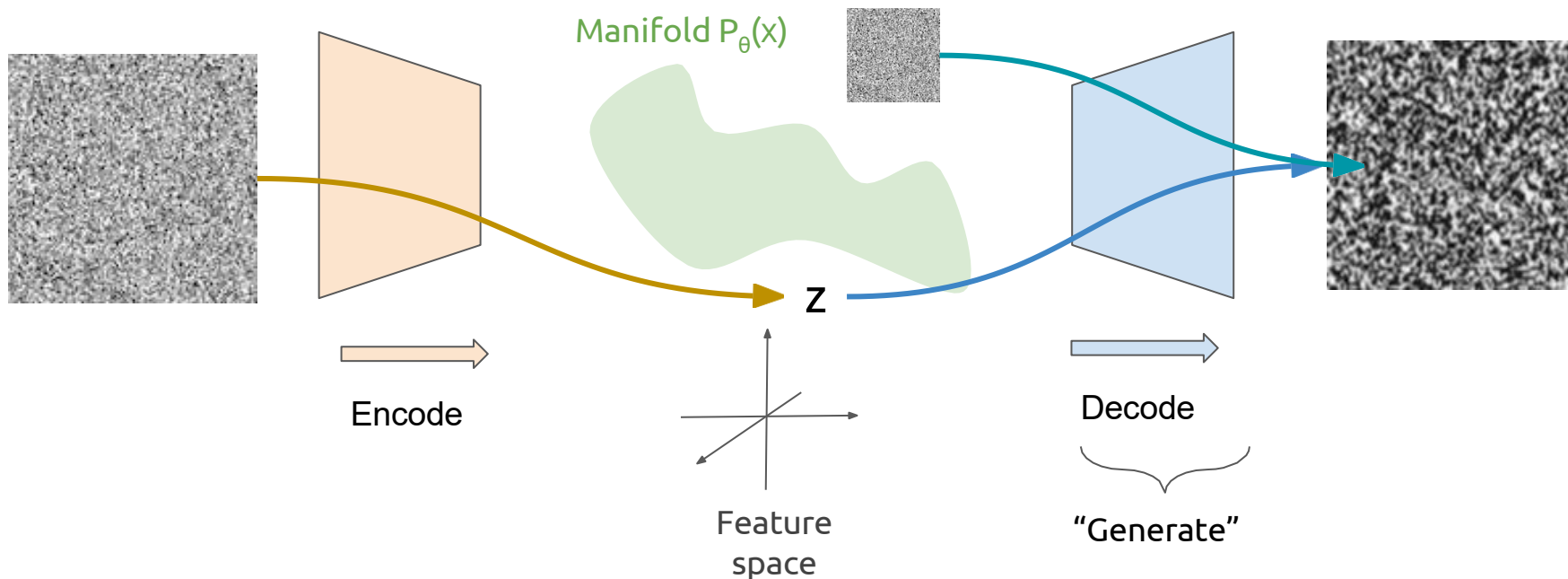
Could we generate new samples by sampling from a normal distribution and feeding it into the encoder, or the decoder (as in GANs) ?



Auto-Encoder (AE)

Could we generate new samples by sampling from a normal distribution and feeding it into the encoder, or the decoder (as in GANs) ?

No, because the noise (or encoded noise) would be out of the learned manifold.

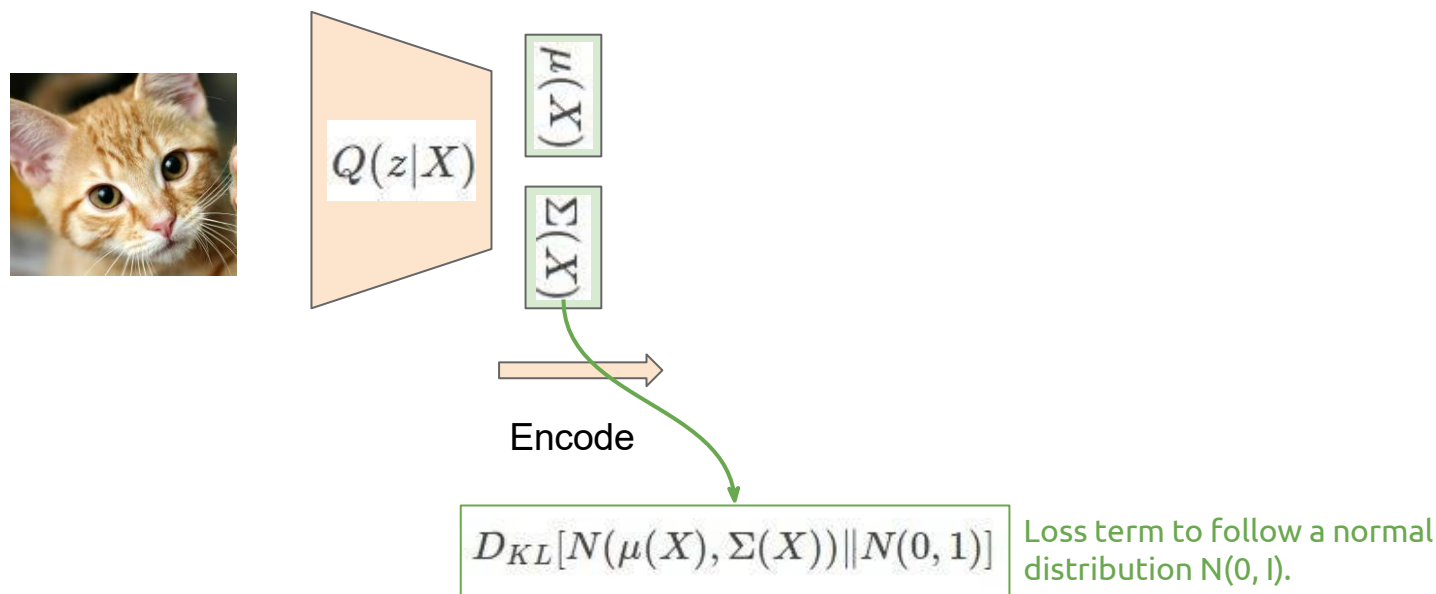


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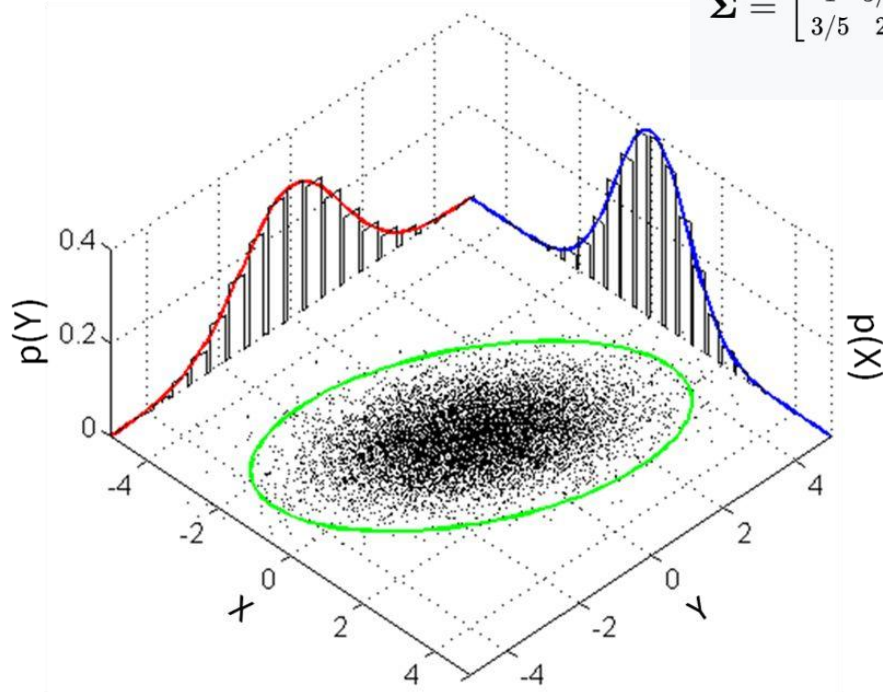
Variational Auto-Encoder (AE)

Encoder: Predict the mean $\mu(x)$ and covariance $\Sigma(x)$ of a multivariate normal distribution.



Maths 101: Multivariate normal distribution

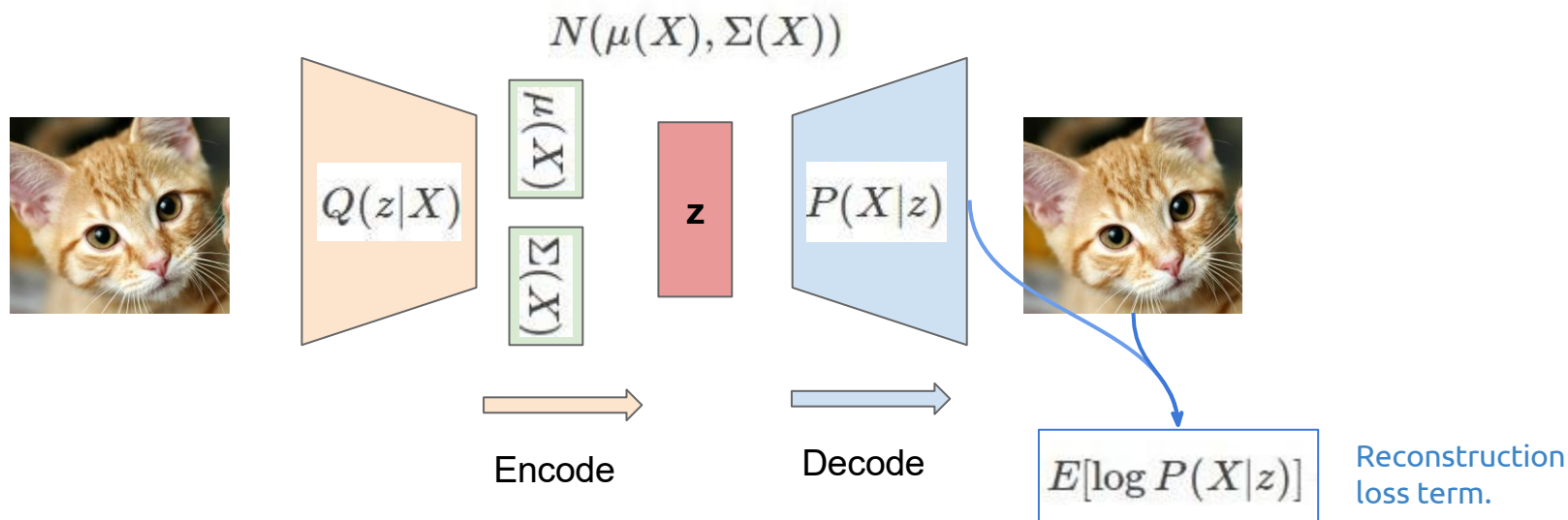
Many sample points from a multivariate normal distribution with $\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix}$, shown along with the 3-sigma ellipse, the two marginal distributions, and the two 1-d histograms.



Notation	$\mathcal{N}(\mu, \Sigma)$
Parameters	$\mu \in \mathbf{R}^k$ — location $\Sigma \in \mathbf{R}^{k \times k}$ — covariance

Variational Auto-Encoder (AE)

Decoder: Trained to reconstruct the input data from a z sampled from $N(\mu, \Sigma)$.



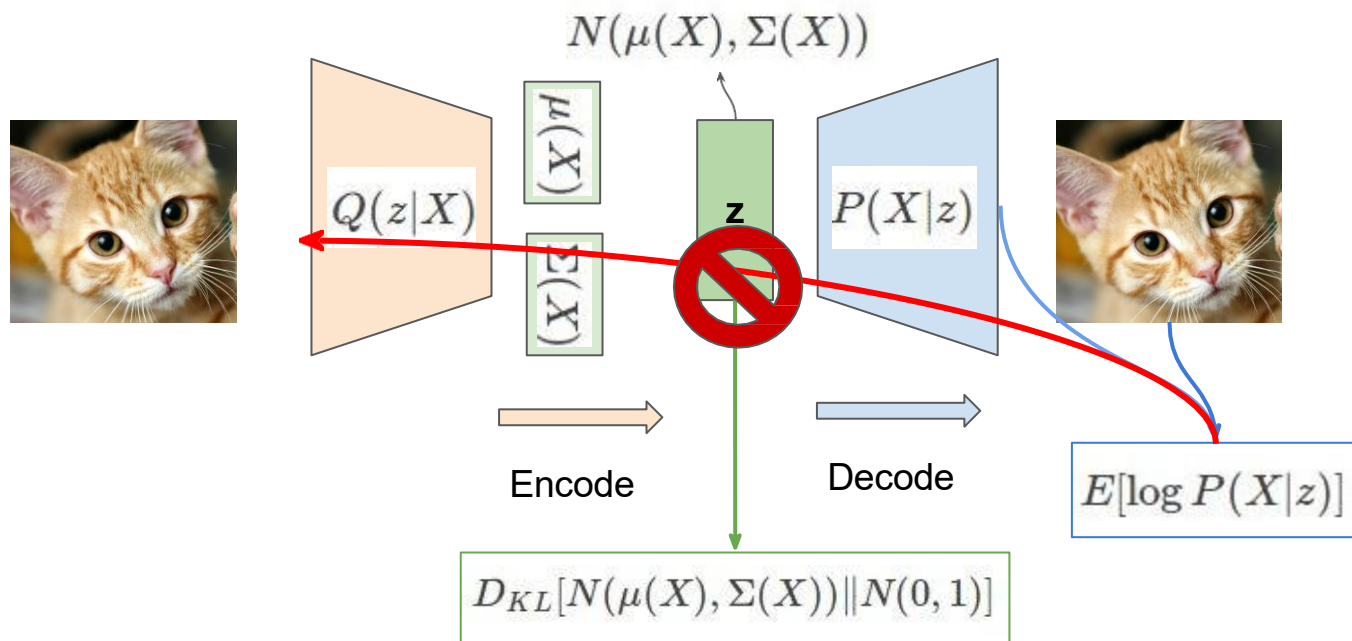
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Reparametrization Trick

Challenge:

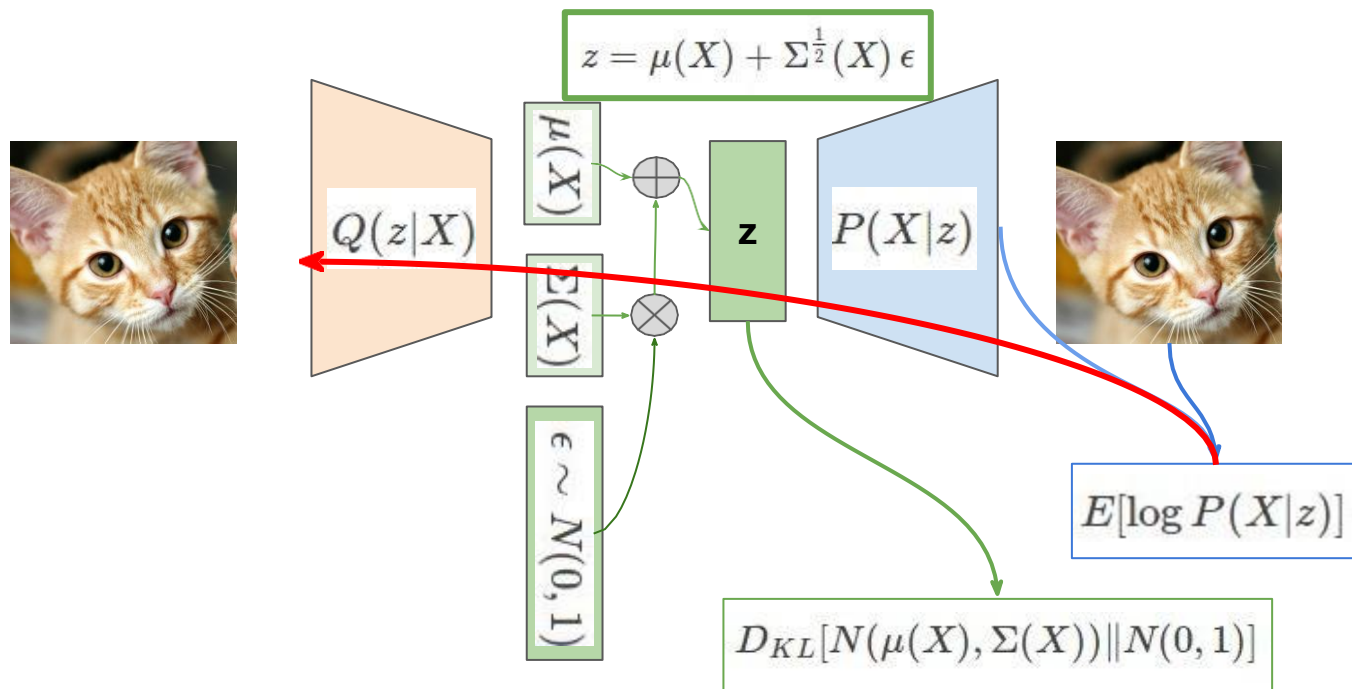
We cannot backprop through sampling of $N(\mu(X), \Sigma(X))$ because **"Sampling" is not differentiable!**



Reparametrization Trick

Solution: Reparameterization trick

Sample $\epsilon \sim N(0, 1)$ and define $\mathbf{z}$ from it, multiplying by $\Sigma(X)$ and summing $\mu(X)$

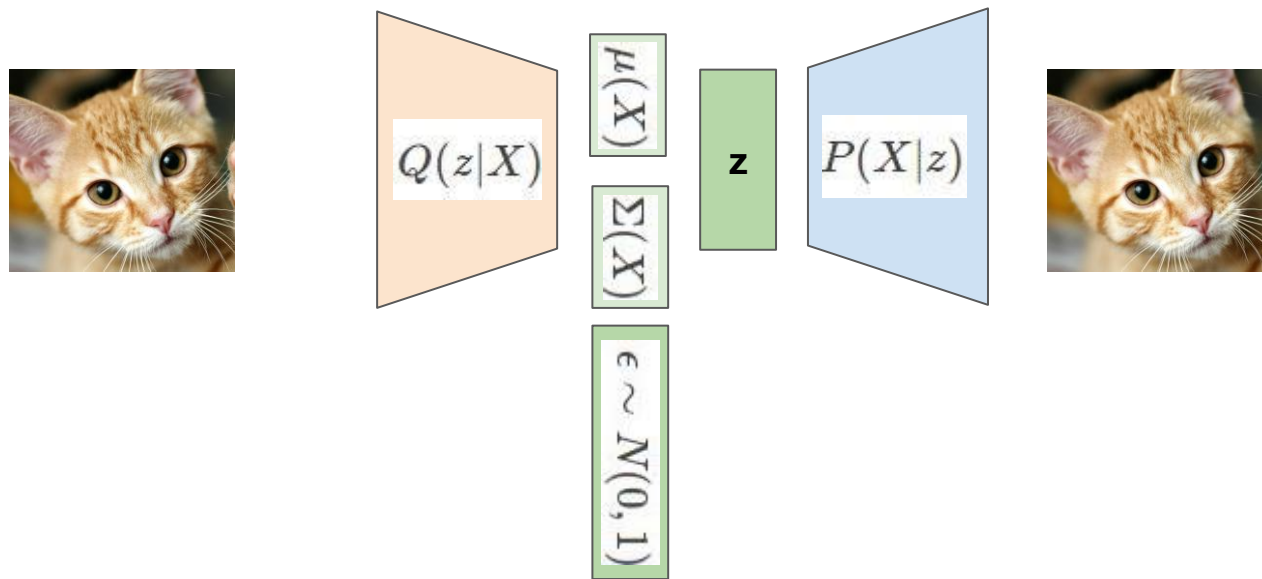


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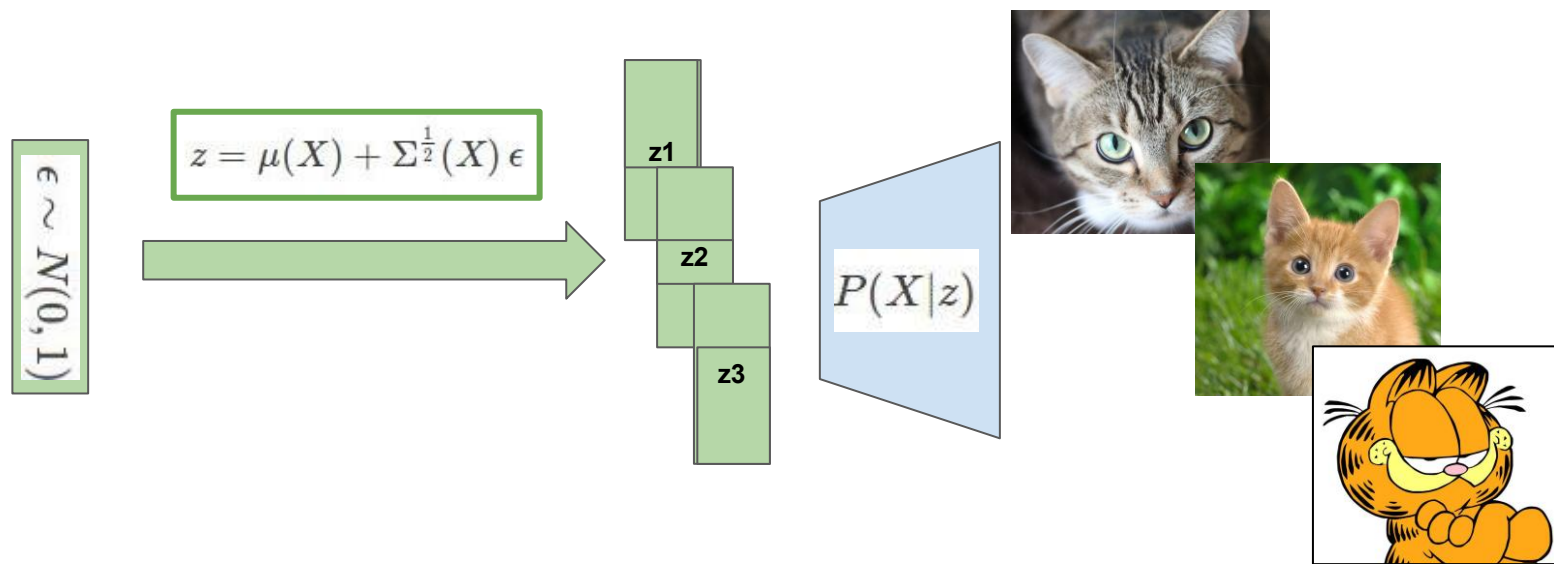
Generative behaviour

How can we now generate new samples once the underlying generating distribution is learned?



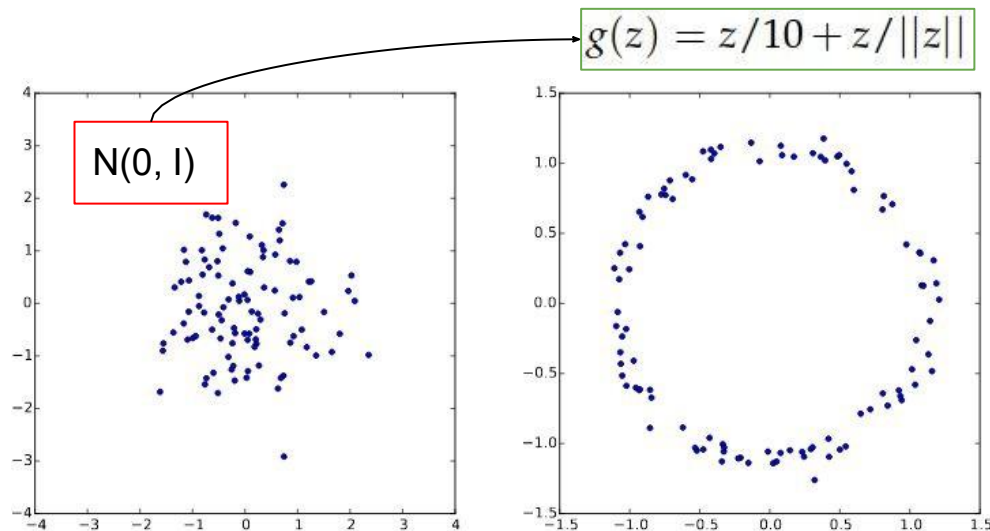
Generative behaviour

We can sample from our prior $N(0, I)$, discarding the encoder path.

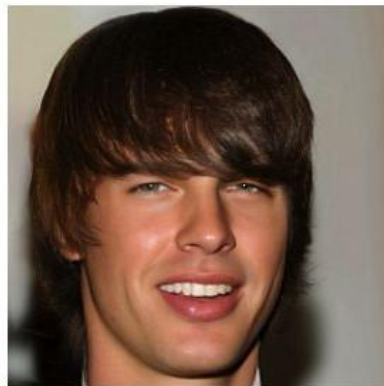


Generative behaviour

Example: $P(x)$ can be modelled mapping a simple normal distribution $N(0, I)$ through a powerful non-linear function $g(z)$.



Generative behaviour

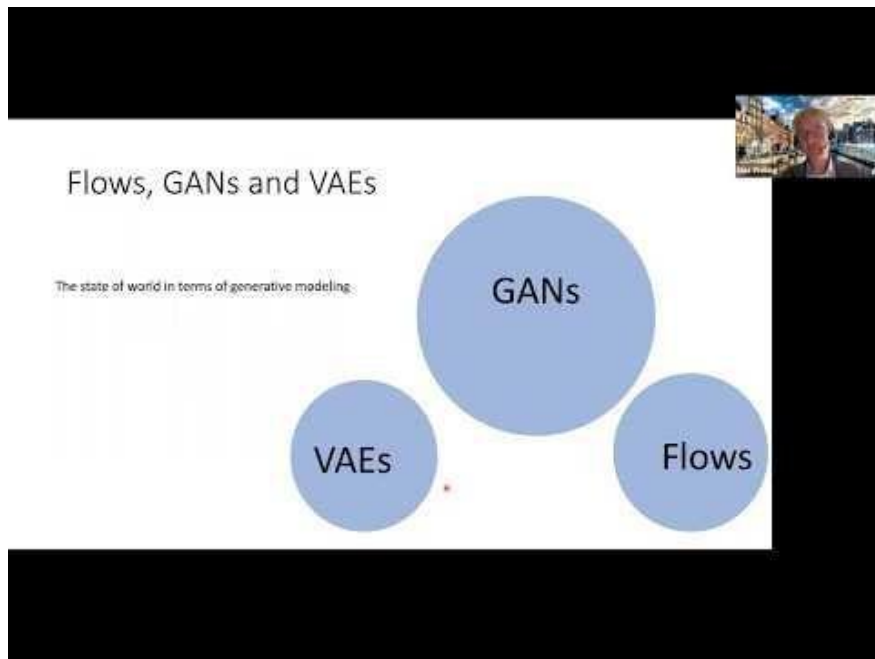


Generative behaviour

Walking around $\mathbf{z}$ manifold dimensions gives us spontaneous generation of samples with different shapes, poses, identities, lightning, etc..



Learn more about VAEs



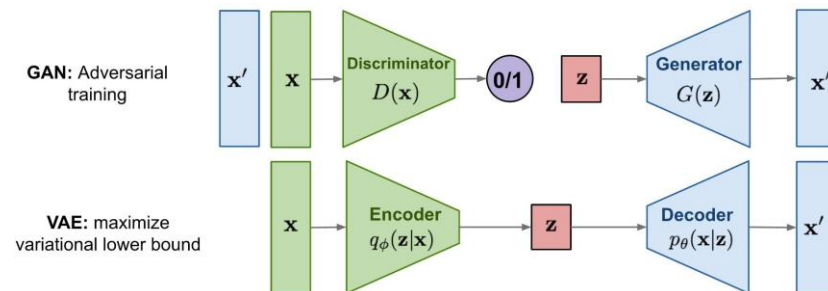
Max Welling - University of Amsterdam (2020)



Andriy Mnih (UCL - Deepmind 2020)

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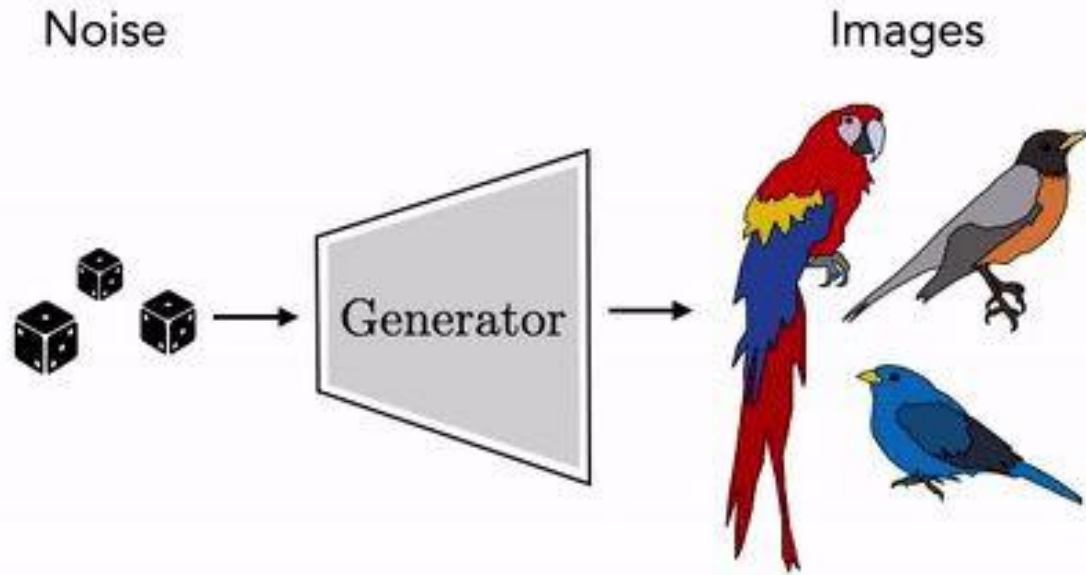
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 - **Forward diffusion process**
 - Reverse denoising process
 - Auto-regressive

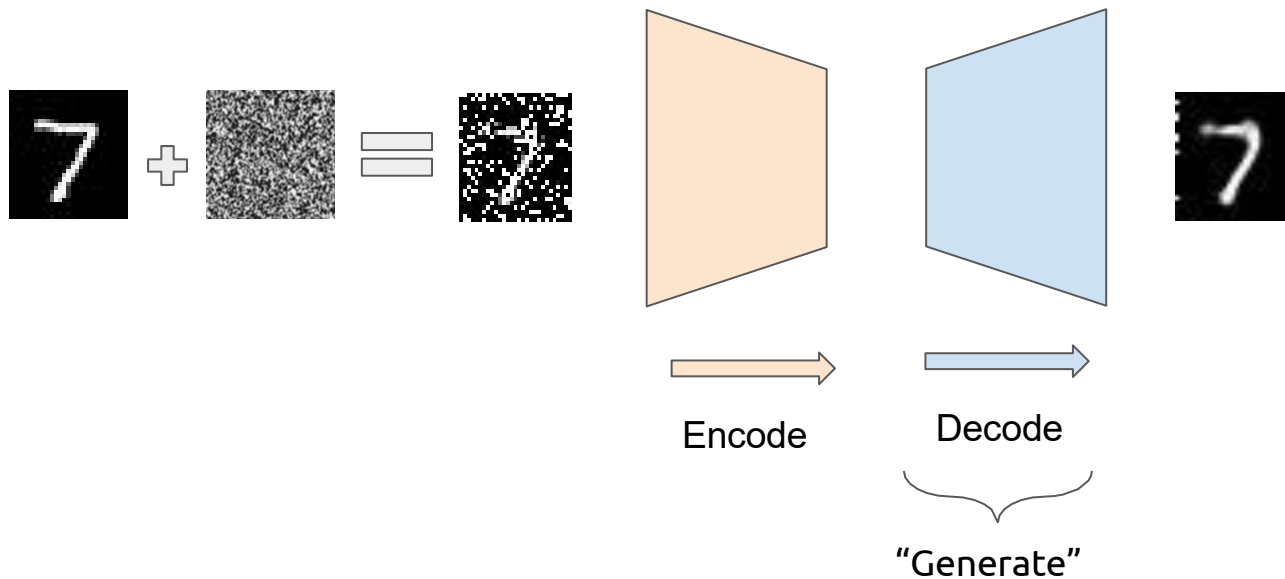
Forward Diffusion Process



Outline

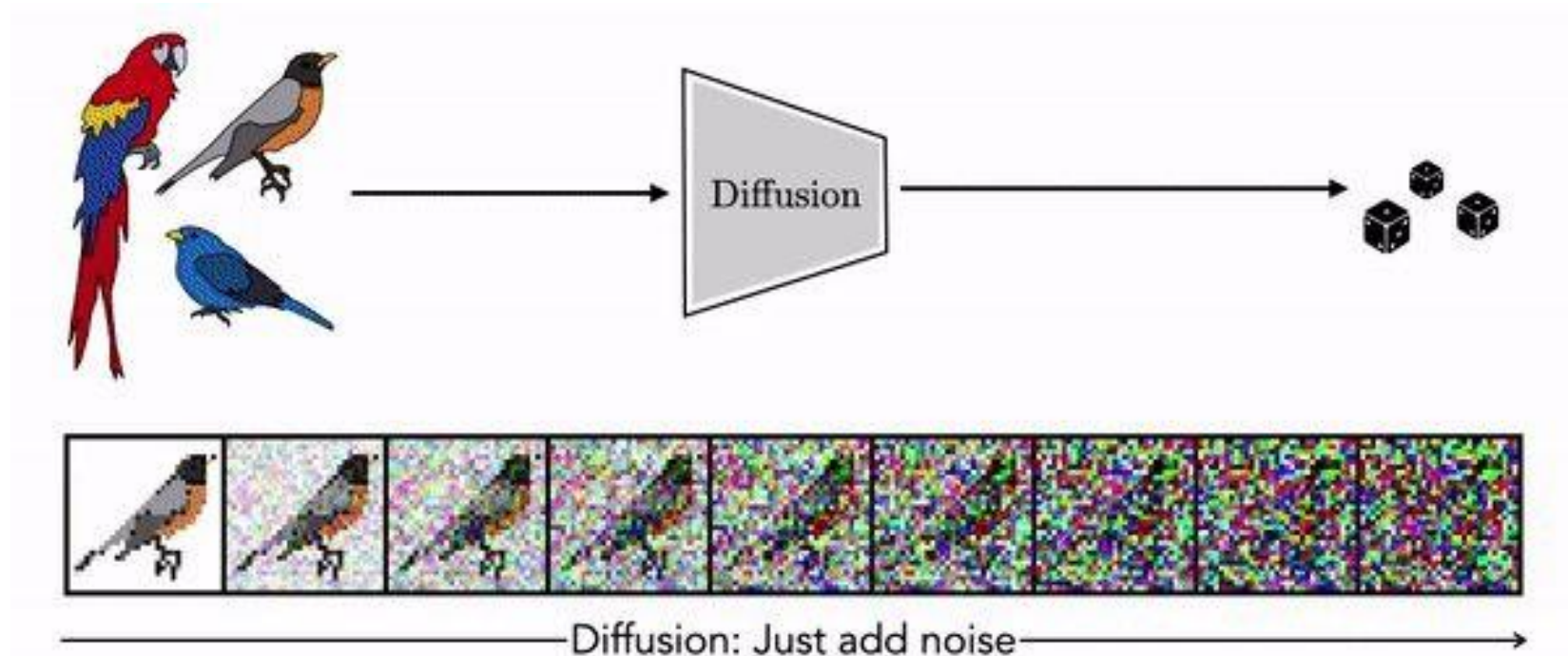
1. Motivation
2. Discriminative vs Generative Models
3. Sampling
- 4. Architectures**
 - Generative Adversarial Networks (GANs)
 - Variational Autoencoders (VAEs)
 - **Denoising Diffusion Models (DDM)**
 - Forward diffusion process
 - **Reverse denoising process**
 - Auto-regressive

Denoising Autoencoder (DAE)

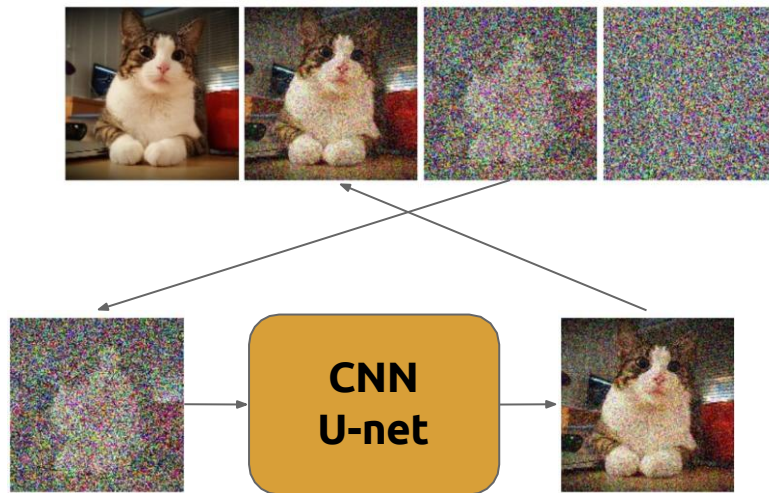
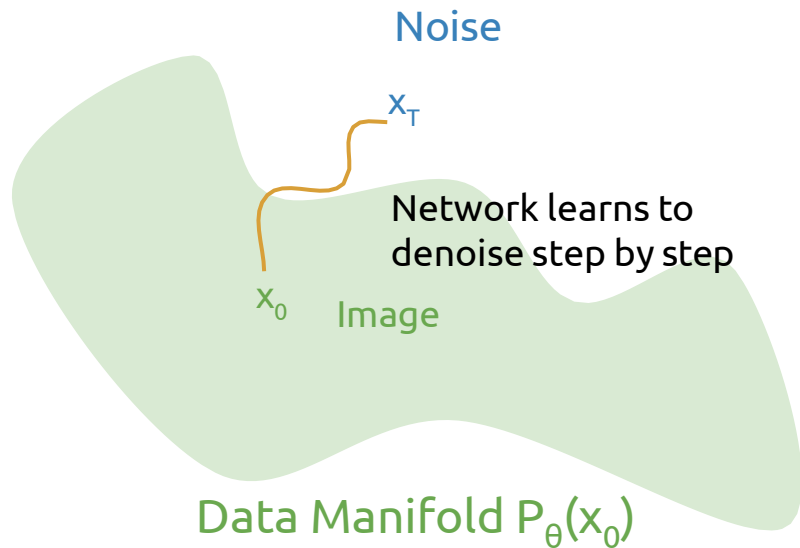


#DAE Vincent, Pascal, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. ["Extracting and composing robust features with denoising autoencoders."](#) ICML 2008.

Reverse Denoising process



Reverse Denoising process



What is the dimension of the latent variable in diffusion models ?

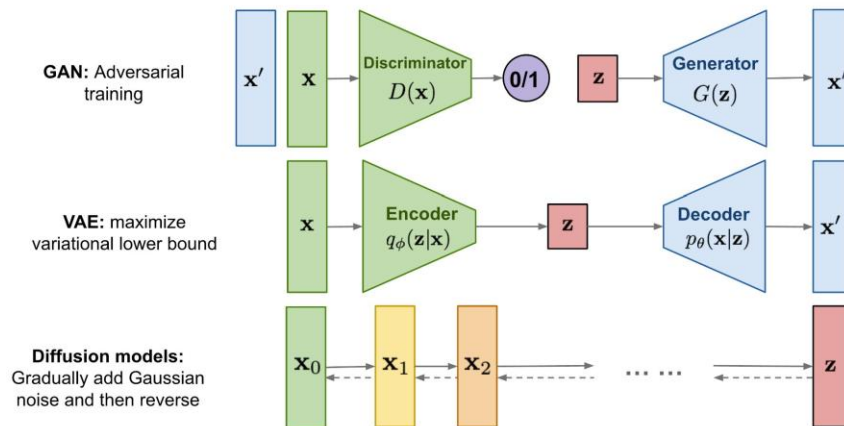
Same dimensionality as the diffused data.

Outline

1. Motivation
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4. Architectures

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Outline

1. Motivation
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 - Generative Adversarial Networks (GANs)
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 - Denoising Diffusion Models (DDM)
 - **Auto-regressive Models (AR)**

Motivation



Dustin Tran

@dustinvtran

Senior Research Scientist at Google Brain. AI, Bayesian statistics, machine learning.

Seguint



Yann LeCun

@ylecun

Professor at NYU. Chief AI Scientist at Meta. Researcher in AI, Machine Learning, Robotics, etc. ACM Turing Award Laureate.

Seguint



Yann LeCun @ylecun · 14 de maig

Researchers in speech recognition, computer vision, and natural language processing in the 2000s were obsessed with accurate representations of uncertainty.

1/N



26



274



1.694



Dustin Tran

@dustinvtran

En resposta a [@ylecun](#)

I'd say probabilistic generative models did win out in the end. Autoregressive models $p(x_{\text{new}}|x_{\text{prev}})$ are the universal API for all tasks in text, and increasingly so in other domains.

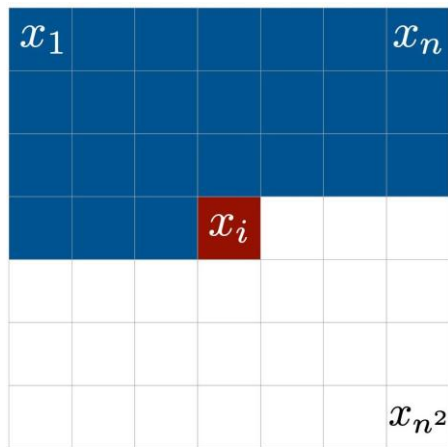
[Tradueix el tuit](#)

1:23 a. m. · 15 de maig de 2022 · Twitter Web App

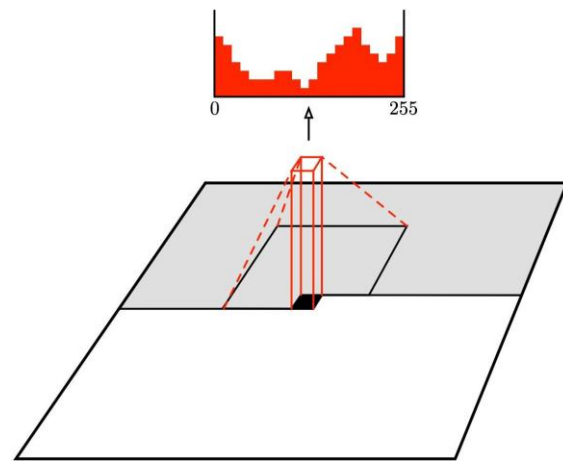
PixelRNN

An RNN predicts the probability of each sample x_i with a categorical output distribution: Softmax

$$p(\mathbf{x}) = \prod_{i=1}^{n^2} p(x_i | x_1, \dots, x_{i-1})$$



Context



PixelRNN

Why are not all completions identical ?
(aka how can AR offer a generative behaviour ?)

occluded

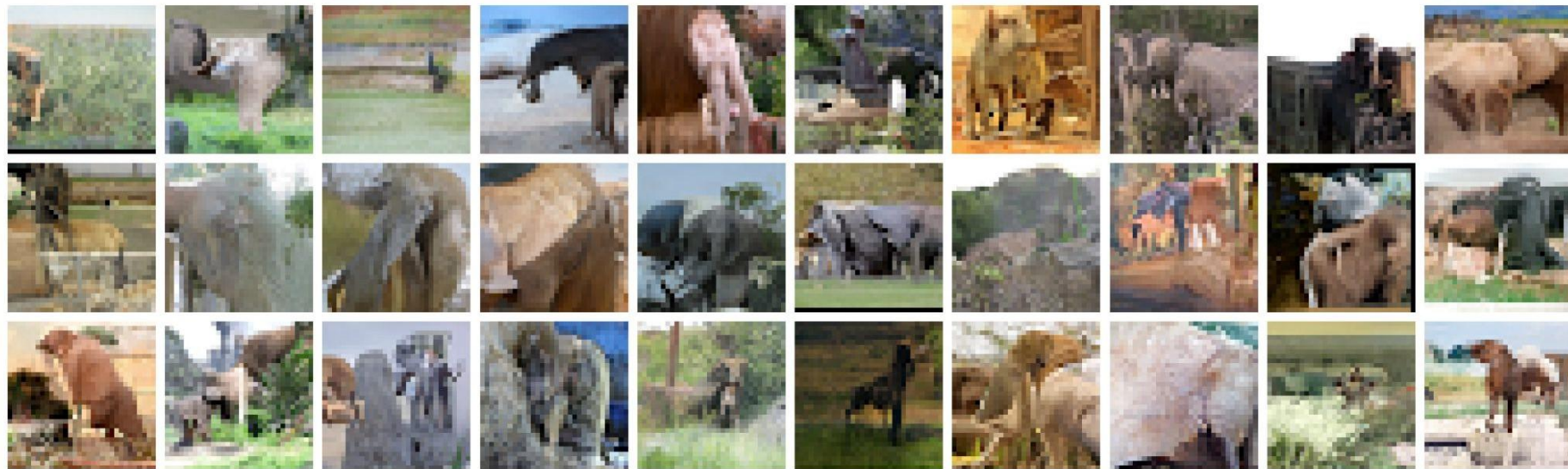


completions



original

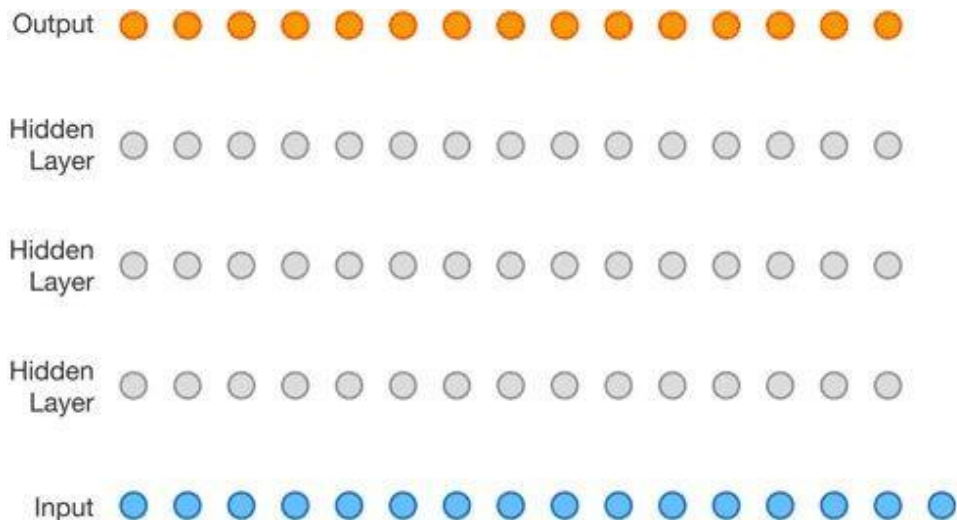
PixelCNN



African elephant

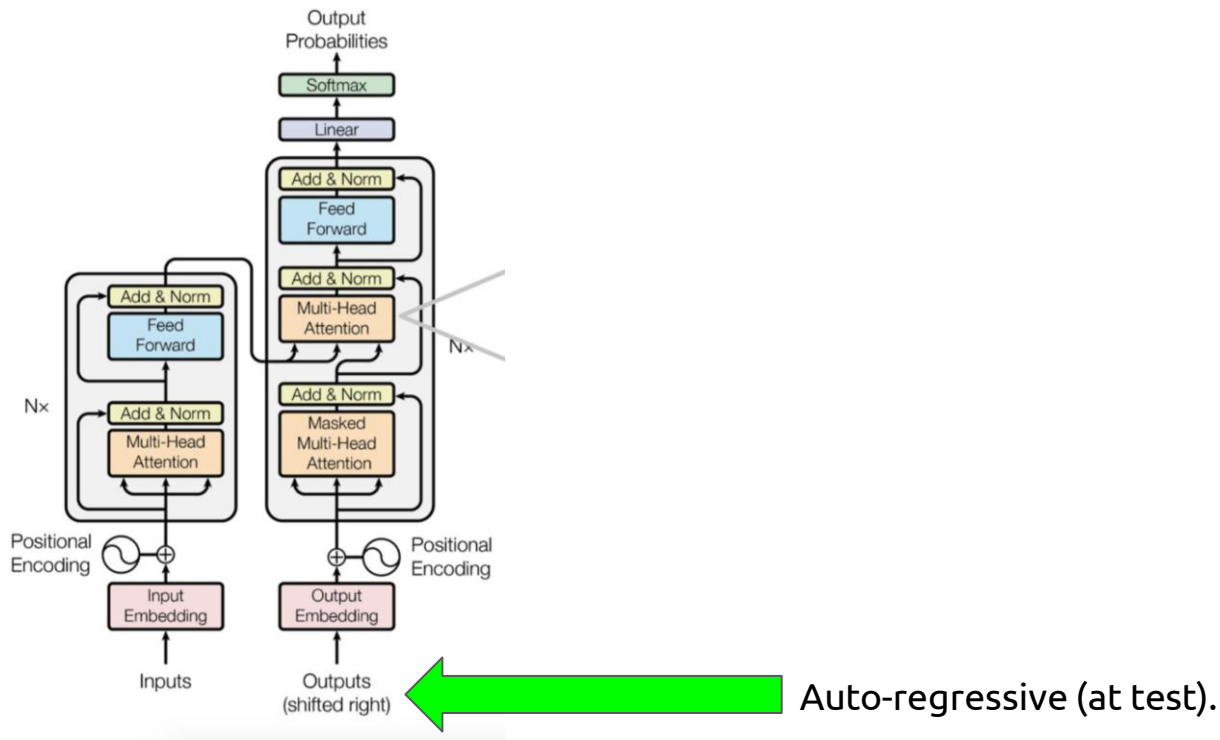
Wavenet

Wavenet used dilated convolutions to produce synthetic audio, sample by sample, conditioned over by receptive field of size T :



#Wavenet Oord, Aaron van den, Sander Dieleman, Heiga Zen, Karen Simonyan, Oriol Vinyals, Alex Graves, Nal Kalchbrenner, Andrew Senior, and Koray Kavukcuoglu. ["Wavenet: A generative model for raw audio."](#) arxiv 2016. [\[blog\]](#)

The Transformer



#Transformer Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I.. [Attention is all you need](#). NeurIPS [ustrated Transformer](#)" (2018)

The Transformer

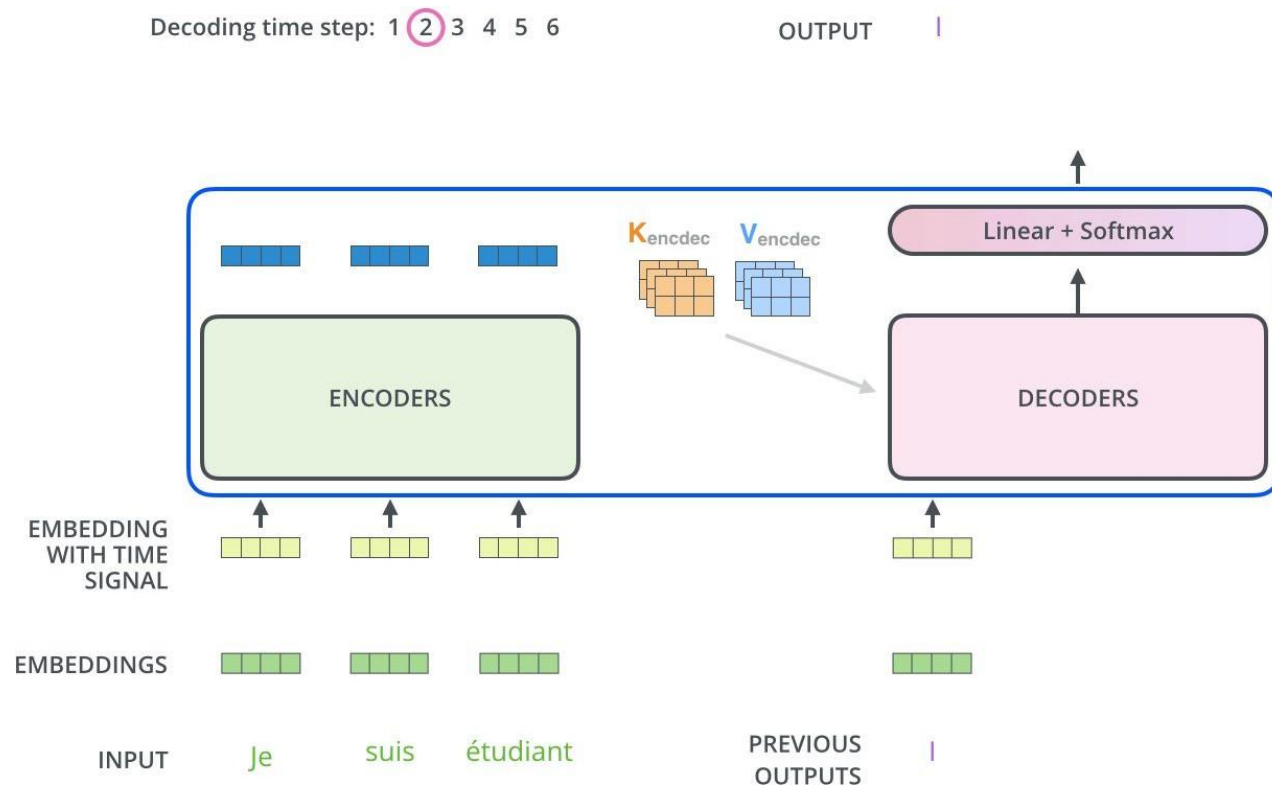


Figure: Jay Alammar, [“The illustrated Transformer”](#) (2018)

Text completion

“GPT-2 is trained with a simple objective: predict the next word, given all of the previous words within some text.”

Condition

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

Generated completions

The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Zero-shot learning

GPT-2/3 can also solve tasks for which it was not trained for (zero-shot learning).

Text

The 2008 Summer Olympics torch relay was run from March 24 until August 8, 2008, prior to the 2008 Summer Olympics, with the theme of “one world, one dream”. Plans for the relay were announced on April 26, 2007, in Beijing, China. The relay, also called by the organizers as the “Journey of Harmony”, lasted 129 days and carried the torch 137,000 km (85,000 mi) – the longest distance of any Olympic torch relay since the tradition was started ahead of the 1936 Summer Olympics.

After being lit at the birthplace of the Olympic Games in Olympia, Greece on March 24, the torch traveled to the Panathinaiko Stadium in Athens, and then to Beijing, arriving on March 31. From Beijing, the torch was following a route passing through six continents. The torch has visited cities along the Silk Road, symbolizing ancient links between China and the rest of the world. The relay also included an ascent with the flame to the top of Mount Everest on the border of Nepal and Tibet, China from the Chinese side, which was closed specially for the event.

Reading Comprehension

Q: What was the theme?

A: “one world, one dream”.

Q: What was the length of the race?

A: 137,000 km

Q: Was it larger than previous ones?

A: No

Q: Where did the race begin?

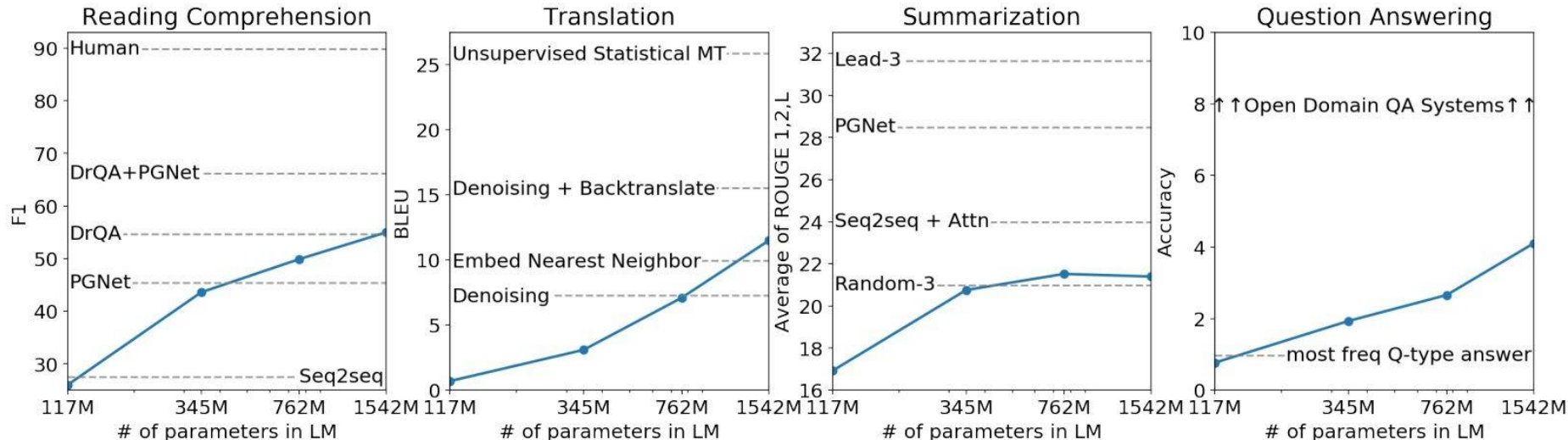
A: Olympia, Greece

#GPT-2 Alec Radford, Jeffrey Wu, Dario Amodei, Daniela Amodei, Jack Clark, Miles Brundage, Ilya Sutskever, [“Better Language Models and Their Implications”](#). OpenAI Blog 2019.

Zero-shot learning

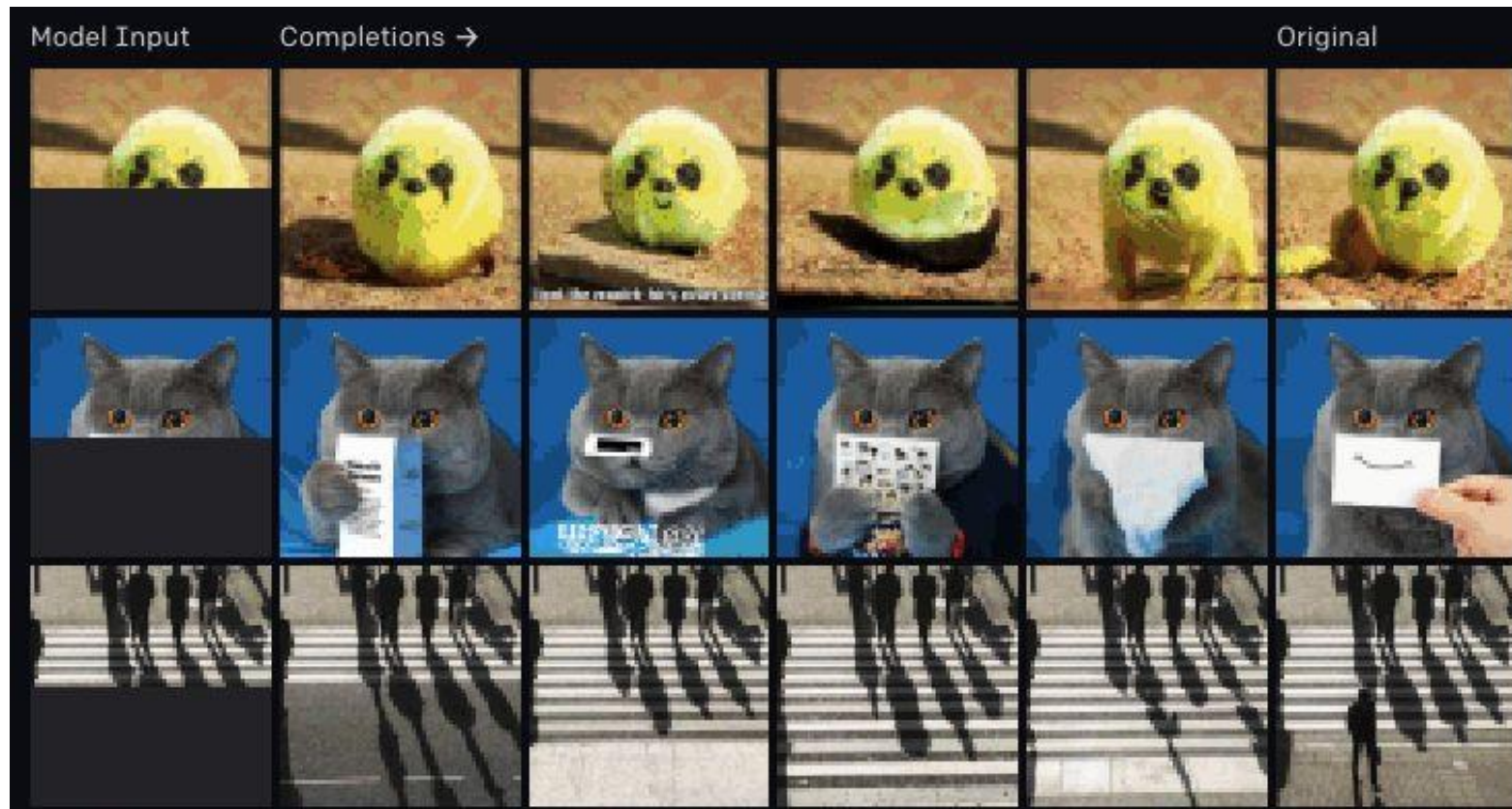
“GPT-2 is trained with a simple objective: predict the next word, given all of the previous words within some text.”

Zero-shot task performances (GPT-2 was never trained for these tasks)



#GPT-2 Alec Radford, Jeffrey Wu, Dario Amodei, Daniela Amodei, Jack Clark, Miles Brundage, Ilya Sutskever, [“Better Language Models and Their Implications”](#). OpenAI Blog 2019.

GPT-2 / GPT-3



#iGPT Chen, M., Radford, A., Child, R., Wu, J., Jun, H., Luan, D., & Sutskever, I. [Generative Pretraining from Pixels](#). ICML 2020.

ChatGPT / GPT-4

Step 1

Collect demonstration data and train a supervised policy.

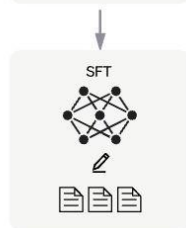
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3.5 with supervised learning.



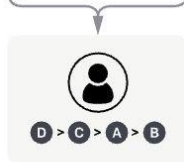
Step 2

Collect comparison data and train a reward model.

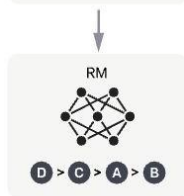
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



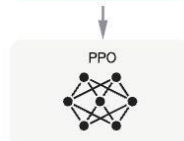
Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

A new prompt is sampled from the dataset.



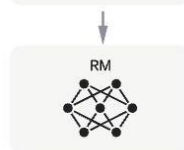
The PPO model is initialized from the supervised policy.



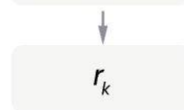
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



#ChatGPT [\[blog\]](#)

#GPT-4 (OpenAI) [GPT-4 Technical Report](#). arxiv 2023. [\[blog\]](#)

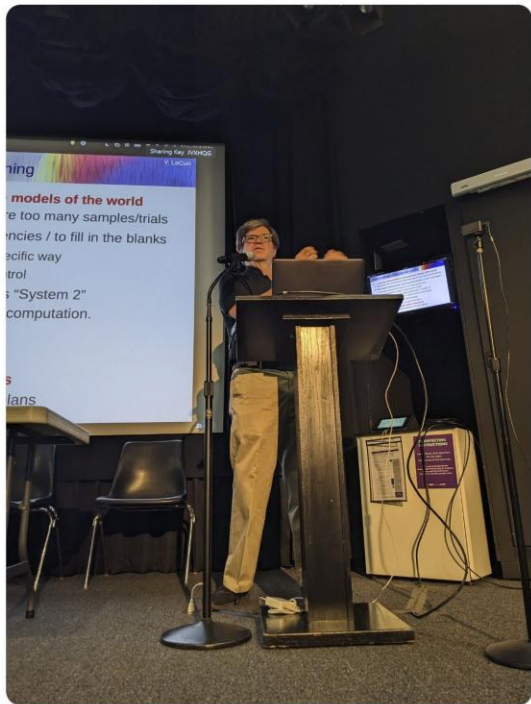
Discussion



Raphaël Millière
@raphaelmilliere

Yann LeCun kicking off the debate with a bold prediction: nobody in their right mind will use autoregressive models 5 years from now
[#phildeeplearning](#)

Tradueix el tuit



10:49 p. m. · 24 de març de 2023 · 560,5m Visualitzacions



Yann LeCun
@ylecun

I have claimed that Auto-Regressive LLMs are exponentially diverging diffusion processes.

Here is the argument:

Let e be the probability that any generated token exits the tree of "correct" answers.

Then the probability that an answer of length n is correct is $(1-e)^n$

[Tradueix el tuit](#)

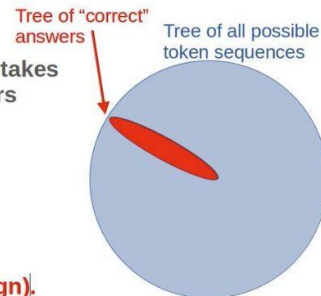
Unpopular Opinion about AR-LLMs

- ▶ Auto-Regressive LLMs are **doomed**.
- ▶ They cannot be made factual, non-toxic, etc.
- ▶ They are not controllable

- ▶ Probability e that any produced token takes us outside of the set of correct answers
- ▶ Probability that answer of length n is correct:

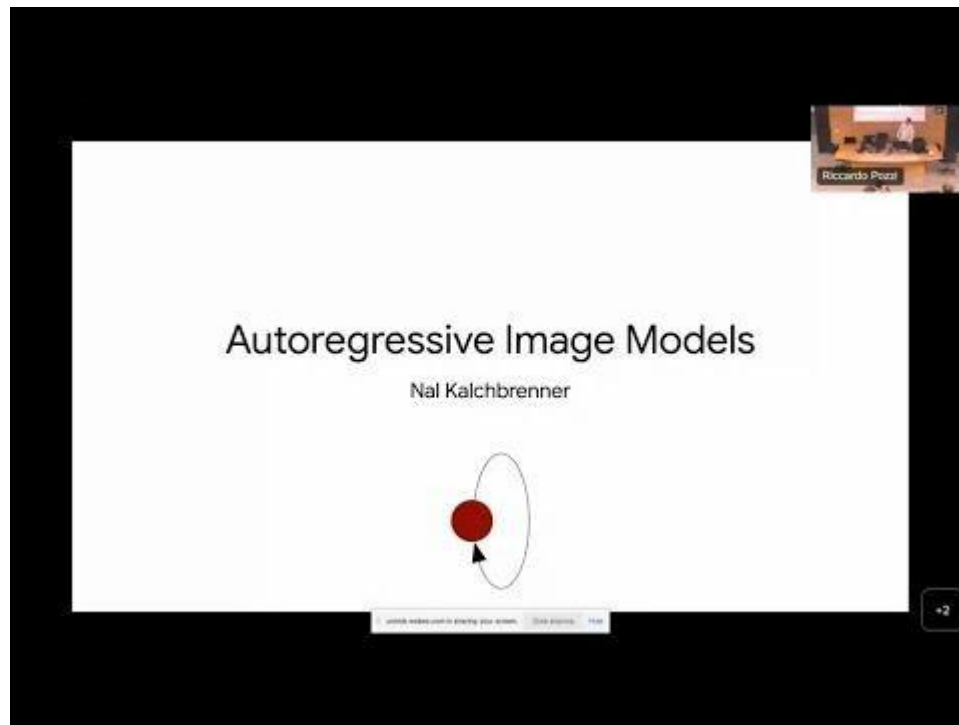
- ▶ $P(\text{correct}) = (1-e)^n$

- ▶ **This diverges exponentially.**
- ▶ **It's not fixable (without a major redesign).**



12:43 a. m. · 27 de març de 2023 · 530,9m Visualitzacions

Learn more about AR models

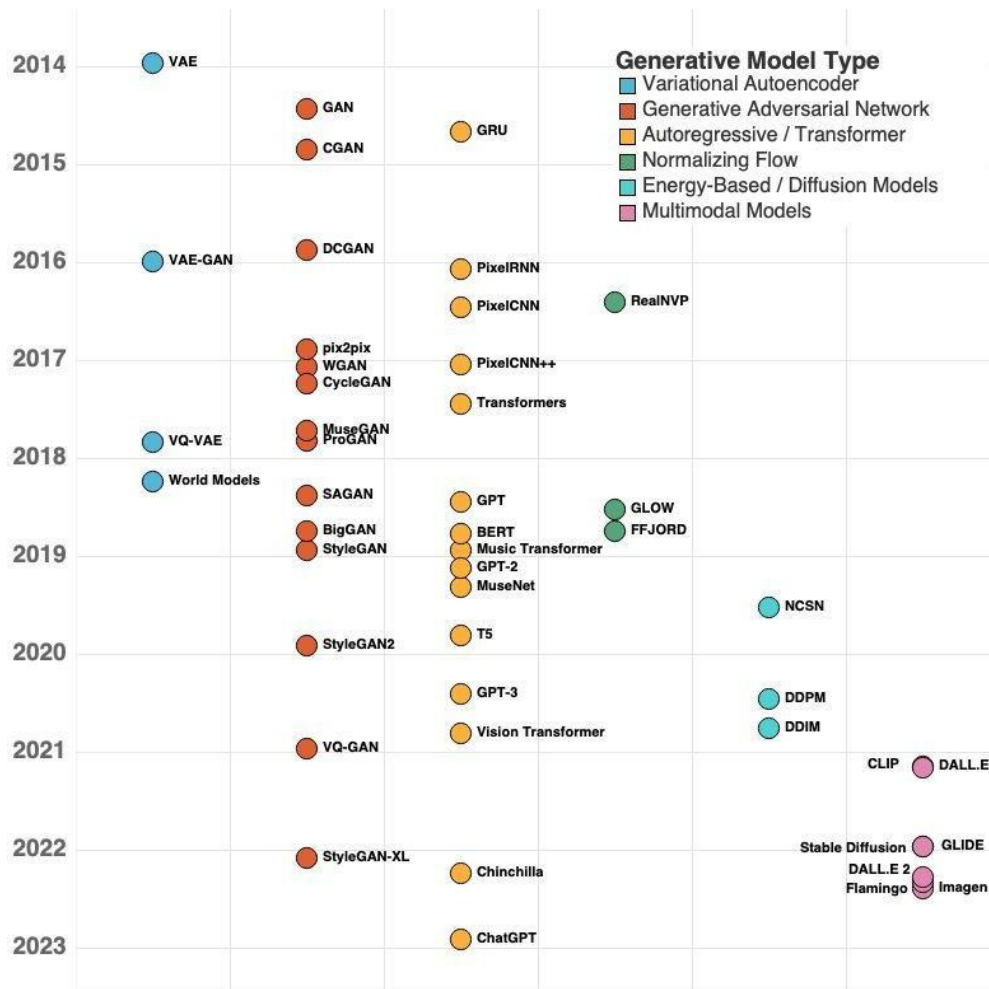


Nal Kalchbrenner, [Mediterranean Machine Learning Summer School 2022](#).

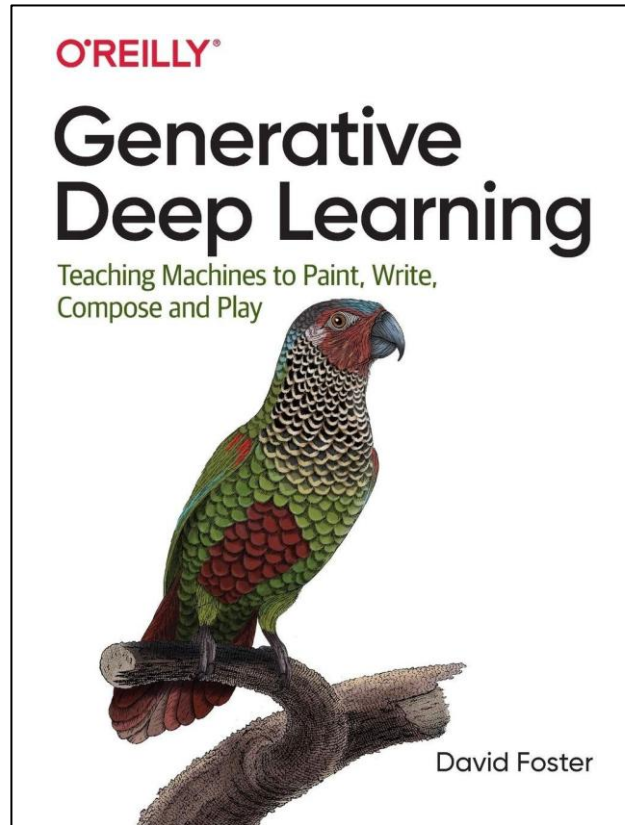
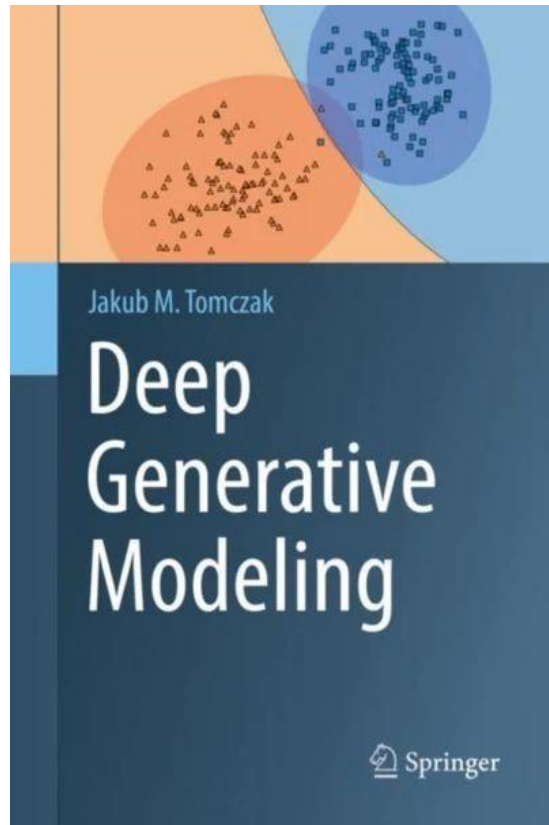
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Generative AI Timeline



Recommended books



Interview of David Foster for
Machine Learning Street Talk
(2023)